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Medical Robotics

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Chapter 1

Introduction

This chapter provides an overview of the medical robotics area, including key research and development challenges [Dom03]. The work developed by our research group at the Institute of Systems and Robotics - University of Coimbra is described, including remote ultrasound diagnosis [SC14a], [SC15], [SC18], [CC20] and laparoscopic and beating-heart surgeries [DC15], [DC14], [CD17].

Medical robotics areas

Patient-centered robotics

- ▶ Assistive robotics. Robots and mechatronic devices which improve the quality of life of disabled people, through increased personal independence
- ▶ Rehabilitation robotics. Robots and mechatronic devices for clinical therapy in neuro-motor rehabilitation

Physician-centered robotics

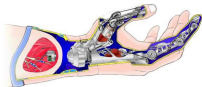
- ▶ Non-surgical robotics. Robots for diagnosis
- ▶ Surgical robotics. Telemanipulation and comanipulation surgeries

Artificial Intelligence (AI)

- ▶ Intelligent robotics boosted by Large Language Models (LLMs), Physical AI, Edge AI, and Human-AI interaction

Assistive robotics

Prosthetic devices



Smart living spaces



Functional electrical stimulation (FES)



Robotic aids



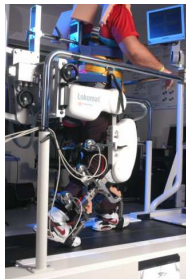
Personal assistants



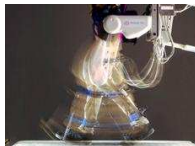
Orthotic devices



Rehabilitation robotics



Active / passive rehabilitation



R & D issues

Assistive robotics

- ▶ Robust indoor navigation system
- ▶ Natural language interfaces and adaptation to patient
- ▶ Miniature and wearable devices to measure physiological parameters
- ▶ Human movement / behavior interpretation (including speech, facial expression and gestures)

Rehabilitation robotics

- ▶ Better human-robot interfaces
- ▶ Haptic interaction
- ▶ Active/passive robot control design

Non-surgical robotics

Robots for diagnosis

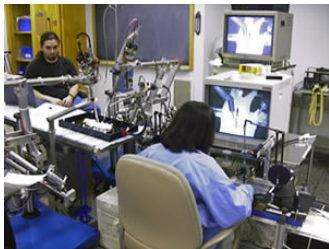


R&D issues

- ▶ Robotic telemanipulation with random time delays (stability issues)
- ▶ Haptic feedback and intuitive HMI interfaces
- ▶ Compliant control [SC14a], [SC15], [SC18]
- ▶ Medical imaging processing for robot navigation [CC20]

Surgical robotics

Telemanipulation and comanipulation [GPG⁺14]



Manual procedures: "machining" rigid surfaces

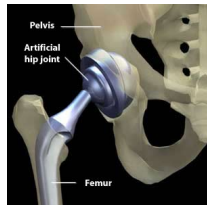
► Applications

- Total/partial knee arthroplasty
- Total hip arthroplasty



► Problems

- Complex geometry
- Localization of cutting planes and drilling axes
- Accuracy
- Force control
- Cutting, milling and drilling



Manual procedures: insertion of instrument in soft tissues

► Applications

- Biopsy, radio frequency ablation of tumors, cryotherapy
- Wide use in neurosurgery, cardiac surgery, urology and abdominal surgery
- Image-guided (CT-scan, MRI) alignment and insertion

► Problems

- The surgeon is exposed to radiation
- Mental registration of the patient's anatomy to the image in targeting
- Precise hand-eye coordination
- Force control during insertion while penetrating tissues with heterogeneous stiffness
- Physiological motion compensation
- Avoiding vital areas



Manual procedures: minimally-invasive surgery (MIS)

► Applications

- Laparoscopic and beating-heart surgeries [DC15], [DC14], [CD17]

► Problems

- Monocular vision - depth information lost
- Hand-eye coordination (fulcrum effect)
- Three hands required
- No tactile feedback
- Comfort of the surgeon
- No force feedback (friction on the trocar)
- Loss of internal mobility due to kinematics constraints induced by the trocar
- Restricted workspace
- Occlusion of the field of view
- Physiological motion compensation
- Critical area avoidance



How can robotics help surgery? I

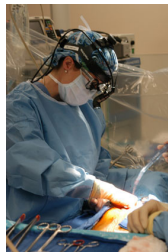
Surgeon

► Advantages

- Efficient natural sensors
- Dexterity coordination
- Capacity in reasoning and learning
- Skill adaptation

► Problems

- Subject to fatigue
- Stability
- Precision
- Unable to see through tissues
- Subject to radiations



How can robotics help surgery? II

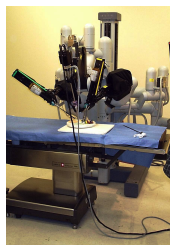
Robot

► Advantages

- Geometric accuracy
- Precision in controlling forces
- Possibility to work in hostile environment
- Repeatability
- No fatigue
- Stationarity
- Rapidity

► Problems

- Weak capacity in deciding, learning and adapting
- Incomplete models
- Reliability



How can robotics help surgery? III

Surgeon + Robot

► Advantages

- Merging synergies to enhance surgical performance
- Better accuracy
- Trauma decreased
- Post operative comfort and fast recovering
- Compensation for surgeon's hand tremor
- Motion and force augmentation or scaling (e.g., for microsurgery)
- Better comfort for the surgeon

► Problems

- Too bulky; cost effectiveness; technical team in the OR; clinical added value: not yet clear (lack of benchmarks); user friendly interfaces; safety



Advantages of surgical robotics

Difficulties that a robotic system can help to solve

- ▶ Precise localization (position and orientation) of instruments with reference to pre-operative or intra-operative imaging
- ▶ Complex and accurate path following (e.g., milling a bone, targeting a tumor from multiple radiation ports)
- ▶ Cancel the hand-eye coordination problem (e.g., in MIS)
- ▶ Real time integration of intra-operative data: image-guided motion (e.g., needle insertion)
- ▶ Visual-servoing (e.g., to compensate for physiological motions and patient's motions)
- ▶ Force-controlled motion (e.g., machining, skin harvesting)
- ▶ Limitation of risks: haptic walls to move into safe regions
- ▶ Heavy loads (e.g., microscope)
- ▶ 3rd hand

Advantages of surgical robotics

Expected “added-values”

- ▶ In neurosurgery, percutaneous therapy, radiosurgery: limits collateral effects due to lesions of instruments or radiations while accessing smaller and smaller targets closer and closer to vital areas; minimizes the radiation for the patient; removes the operator from hazardous environment such as X-ray
- ▶ In orthopedic: less revision surgeries; longer life expectancy of prostheses; less risk (e.g., pedicular screw placement)
- ▶ In MIS: control of additional degrees of freedom at the distal part; haptic feedback; performing surgeries that cannot be executed manually (e.g., beating heart surgery) still minimizing invasiveness; addressing kinematic constraints; less pain and trauma and shorter recovery time
- ▶ Telesurgery
- ▶ Training / mentoring of young surgeons

Problems of surgical robotics

Constraints related to the use of a robotic system in the OR

- ▶ Benefits need to justify cost/complexity
- ▶ No technical team in the OR
- ▶ Skin-to-skin times longer than conventional techniques
- ▶ Clinical added value is not clear: there are no established methods to relate conventional (non-robotic) techniques that would serve as benchmarks
- ▶ Easy to use and setup: currently longer
- ▶ Compatible with the environment of the OR (cluttered, other electrical devices)
- ▶ Too bulky and lack of user-friendly interfaces
- ▶ Safety
- ▶ Change surgical procedures (research work)

Safety issues

► Problems

- The robot shares its working space with surgical staff and patient
- “Trial & error” or “doing again” motions are not allowed
- Sterilization constraints

► Solutions

- At the design level: redundancy in control and sensing
- Intrinsically safe design to decrease the level of risk by construction (e.g. limit the actuator power to give the instrument just the energy required by the task)
- Use of reliable components
- At the operation level: dedicated human-machine interfaces and adequate rules for use

Safety issues: example of solutions



Kuka robot for medical applications

- ▶ Proximity sensors to reduce robot speed
- ▶ Collision prevention: safe switches to stop the robot and protective foam covering the arm to absorb kinetic energy

Milestones of surgical robotics

- ▶ **1985.** PUMA 260 robot performed neurosurgical biopsies in 22 patients with enhanced precision [KHJH88]
- ▶ **1988.** PUMA 560 robot performed transurethral resection of the prostate [DHCW89]
- ▶ **1989.** Speedy system, developed at Grenoble for neurosurgery, including electrode implantation for thalamus excitation and biopsies [Lav89]
- ▶ **1992.** Integrated Surgical Systems developed ROBODOC for orthopedic surgery used for total hip replacement. More than 10000 patients benefited from this system
- ▶ **1994.** Computer Motion launched AESOP (an acronym for Automated Endoscopic System for Optimal Positioning). This robotic system is guided by voice. This allows the surgeon to use spoken English to precisely maneuver and position a laparoscope. More than 15000 minimally invasive surgeries have been successfully assisted by AESOP in more than 150 hospitals and surgery centers around the world

Milestones of surgical/non-surgical robotics

- ▶ 1998. Computer Motion launched Zeus system for MIS. The surgeon uses familiar open surgery movements while performing endoscopic procedures through tiny 5-mm ports
- ▶ 1998. Vision and robotics laboratory (Bourges, France) performs tele-echography between Bourges and Katmandou
- ▶ 1999. In January 1999, Intuitive Surgical launched the *da Vinci* Surgical System, and in 2000, it became the first robotic surgical system cleared by the FDA for general laparoscopic surgery
- ▶ 2001. Transatlantic robot-assisted telesurgery between New York and Strasbourg
- ▶ 2010. Surgeons at Barnes-Jewish Hospital hit a milestone in January with their 1000th robotic surgery case using the *da Vinci* surgical system
- ▶ 2016. Partly autonomous robotic surgery [SDO⁺16]
- ▶ 2019. Comparative studies between robotic and non-robotic surgeries for rectal cancer surgery were carried out [LZDL19]

Future directions and technical challenges

▶ New clinical procedures

- ▶ Neuroendoscopy for back pain treatment
- ▶ Gastrointestinal diagnosis and therapy
- ▶ Microsurgery
- ▶ Minimally invasive surgery
- ▶ Plastic and reconstructive surgery

▶ Improving existing procedures

- ▶ Lightweight, smaller, simpler and cheaper robots
- ▶ More integrated systems: plug-and-play robots
- ▶ Real cooperation between surgeon and robot (“Hands-on” concept: comanipulation)

▶ Solutions

- ▶ Portable (patient-mounted) robot
- ▶ “Smart” instruments for simple dedicated tasks rather than multi-functions
- ▶ Intra-body robots: active catheters, “pills” and endoscopes, mobile robots and microtechnologies

Chapter 2

Safety & design

This chapter addresses software and hardware technologies for robot safety and design [Pie03]. Safety [MKJ⁺11], fault tolerant analysis [GV02], real-time computation [Kop11] and degrees-of-freedom design [MS10] are discussed.

Industrial versus medical robots

▶ Industrial robots

- ▶ Isolated from workers (appropriate training to interact with them). Machine workspace without humans
- ▶ Possibility to enter in the workspace (for maintenance purpose or learning procedures) without stopping the machine by activating manual mode

▶ Medical robots

- ▶ Medical robots cooperate with humans (surgeon and staff) and interact with patients
- ▶ Harsh constraints especially for active medical devices
- ▶ Human factors and clinical constraints affect robot design

Human factors

▶ Patient factors

- ▶ Change in working conditions with each patient (e.g., characteristics of soft tissues, position of the patient on the operating table, size of the body and accessibility of the organs)
- ▶ Task and execution specific to a patient: no “trial/error” nor “doing again” movements

▶ Surgeon and staff factors

- ▶ Dedicated user-friendly HMI (surgeons are not robot specialists): task-oriented, allowing an easy manipulation of the system
- ▶ Robot transparency avoiding singularities, mechanical joint limits and reconfiguration procedures
- ▶ Necessity of preoperative studies to plan the intervention
- ▶ Modification of planning during the operation itself, according to the surgeon diagnostic, possible complications or patient organism behavior

Clinical constraints

- ▶ **Sterilization constraints**

- ▶ Usually, the robot is covered by a sterile sleeve while the tool is separately sterilized

- ▶ **Environment constraints**

- ▶ Operating rooms are cluttered with several other medical systems (e.g., radiology, anesthesia and surgery)
- ▶ The robot position with respect to the patient varies between two operations and even a single operation
- ▶ Robot dimensions have to be small

- ▶ **Logistic constraints**

- ▶ The robot has to be easily and quickly transportable in and out of the operating room

- ▶ **Functionality constraints**

- ▶ Medical robots are often designed for specific operations

Safety rules: European directives

- ▶ **ISO 9000 norm** has been modified to comply with the specific constraints of medical devices in the European directive 93/42/CEE.
- ▶ The **EN 46000 certification** enacts the various criteria necessary to classify all the medical devices according to four classes
 - ▶ Class I: Small degree of risk
 - ▶ Class IIa: Medium degree of risk
 - ▶ Class IIb: High potential degree of risk
 - ▶ Class III: Very serious potential of risk
- ▶ **Device classification** depends on
 - ▶ life span use: from a few minutes (temporary) to several years (implantable)
 - ▶ invasiveness or non-invasiveness
 - ▶ surgical or non-surgical use
 - ▶ activeness or inactiveness
 - ▶ vital or non-vital body parts concerned by the device

Surgical robot design

Design rules

- ▶ Redundancy in sensor and control
- ▶ Intrinsically safe components
- ▶ Reliability in design

Along two axis

- ▶ Hardware level (e.g., electrical, mechanical and electromechanical)
- ▶ Software level (e.g., real time operating systems and control algorithms)

Electromechanical concepts of safety I

Intrinsically safe components

- ▶ Limitation of actuator power to satisfy only the required tasks
- ▶ Use of reduction gears such as harmonic drives (high efficiency and low backlash) to reduce robot velocities
 - ▶ However, high reduction ratios imply **non back-drivable structures**, which is problematic for neurosurgery and MIS applications
- ▶ When the robot force becomes too high, a mechanical system enables to quickly drop the tool (e.g., on AESOP, the collar linking the endoscope and the arm is quickly disconnected thanks to a magnetic connection)
- ▶ Joints may also be equipped with mechanical torque limiters mounted on the motor shaft (e.g., on Neurobot or Hippocrate, when a link collides with an obstacle during a motion, it stops moving while the motor shaft still rotates)

Electromechanical concepts of safety II

Intrinsically safe components



- ▶ In case of power breakdown or emergency stop, parking brakes mounted on selected joints prevent the robot from falling down under gravity effect. However, to manually move the arm without actuator control, the brakes have to be released. Many robots tend to vibrate a bit when brakes are applied
- ▶ As an alternative to this solution, gravity compensation may be fulfilled by a passive counterbalancing payload or by a full irreversible structure

Electromechanical concepts of safety III

Redundancy of sensors

- ▶ Increasing information by redundant and independent components decreases hazard rating (e.g., duplication of joint brakes to prevent any breakdown effect, multiple cameras coupled with a Computed Tomography system)
- ▶ Avoid time consuming and potentially hazardous initialization procedures by using a combination of two relative encoders (on motor shaft and joint axis)
- ▶ Examples: two resolvers (Hippocrate, SCALPP), one incremental encoder associated with an absolute encoder (Robodoc), one incremental encoder associated with potentiometer (NeuroSkill robot, SCALPP)
- ▶ Redundancy of sensors are also used for robust state estimation (necessary for state space control)

Electromechanical concepts of safety IV

Mechanical design

- ▶ Avoid the risk of wrenching or cutting wires, by shielding and integrating all leads inside robot links
- ▶ Limiting the working envelope by using mechanical joint limits: physical threshold (+ software threshold)
- ▶ Adapting the number of degrees of freedom (dof) to the required task workspace or use redundant kinematics to avoid collision, increasing dexterity
- ▶ Fit link dimensions to preserve patient and clinical staff safety
- ▶ Avoid numerical or polynomial resolution methods and prefer analytical ones
- ▶ Reject wrist and shoulder singularity configurations out of the workspace as much as possible

Software concepts of safety I

Multi-tasking

- ▶ Design the controller as concurrent processes dedicated to specific tasks including security, Cartesian control, joint control, communication with peripheral units and sensors and HMI communication
- ▶ By tuning the process and variable priorities, an appropriate emergency procedure is switched on as soon as an error is detected. For instance, the dedicated security process may have the higher priority
- ▶ Stable computation time requires closed-form solutions for models
- ▶ Definition of the "best" priority protocol and computation of response times for the worst case scenario [Kop11]. Sampling time analysis to guarantee scalability

Software concepts of safety II

Multiple cores

- ▶ Instead of using several processes, it may be interesting to duplicate the CPU. In this case, each CPU is dedicated to one primitive function: for applications using tracking systems or teleoperated systems for instance, one CPU is dedicated to vision data, a second one to force data and a last one to robot control
- ▶ Process activities are checked by a dedicated watchdog process: at each sampling period, a time data is sent by the other processes. By comparing this data to the last one received, the watchdog can detect if a process is stuck in a blocked status

Software concepts of safety III

System monitoring

- ▶ Together with mechanical joint limits, additional software joint limits have to be implemented. To be effective, these software limits must be lower than the mechanical ones. This fictive redundancy allows to increase the lifetime of mechanical components and to guarantee a limit even if the mechanical joint limit is broken. This feature may also be interesting to dynamically limit the workspace according to the operating mode
- ▶ For systems exerting forces on patients, fixing force limits is mandatory. Also, for needle inserting procedures a software penetration limit (relative to the insertion location) can prevent too deep motions

Software concepts of safety IV

System monitoring

- ▶ Tracking errors at both Cartesian and joint levels enable to detect any control error or mechanical breakdown
- ▶ Operating modes should be thought in such a way that the robot motions can be predicted by the user. For instance, if possible, sequential motions of the axes or smooth path should be preferred
- ▶ As usually with real-time programming, variable time consumer procedures should be avoided to decrease the execution-time of the processes (thus, to increase the sampling frequency, in order to react faster in case of emergency)

Electrical concepts of safety I

Intrinsically safe components

- ▶ Authorize robot motion by using a Dead Man Switch (DMS). It requires that the user exerts a continuous positive action to allow the motion of the arm. By releasing the switch (generally a pedal), the robot is stopped or decelerated.
- ▶ Emergency buttons with a good accessibility must be wired on the robot. Any punched action on this button switches actuator power off. Additional keys on the emergency buttons may be used to authorize only skilled staff to turn on the robot again. With these emergency buttons and a single On/Off switch, power is only authorized when all status checking routines have been completed.

Electrical concepts of safety II

Redundancy

- ▶ Duplication of electrical components in the hardware controller
- ▶ Parallel redundancy
 - ▶ Circuits such as watchdog boards or Input/Output (I/O) components are combined by parallel assembly
- ▶ Serial redundancy
 - ▶ Relays, emergency switches or buttons are usually serial wired

Design

- ▶ Electro-magnetic compatibility should be addressed. Immunity to electro-magnetic fields, electrostatic discharges, and other noise sources

Electrical concepts of safety III

Design

- ▶ Position sensors and actuators should be wired on to different powering lines: when the actuator power is turned off by an emergency procedure, encoders must still be powered. Therefore, joint positions are known at any time, even when motors are uncontrolled, like in a manual procedure, when the robot is manually driven by the surgeon. Another advantage is that unnecessary initialization procedures are avoided when power is switched on again
- ▶ Actuators can be current (or voltage) limited according to the current task by an adjusted power circuit
- ▶ A particular care must be taken to ergonomics. Clear error messages or warning signals (such as specific lights or/and sounds) must warn the user of the system status and errors

Redundancy vs. safety

- ▶ Redundancy in the system architecture, at mechanical, electrical and software levels increases the system complexity
- ▶ System cost increases (e.g., duplication of sensing components)
- ▶ Software sophistication increases (e.g., multiple process interaction running in multiple cores)
- ▶ Is there a trade-off between redundancy and reliability?
- ▶ Multi-criterion approach to guarantee safety without systematically adding local redundancy. Safety components (hardware and software) interact to provide the required safety on a specific function at the system level

Multi-criterion approach

Example

- ▶ Consider a force controlled arm where the contact between the robot end-effector and the patient is monitored by a F/T sensor. Excessive applied force is monitored by
 - ▶ The threshold software controller
 - ▶ A threshold built into the F/T sensor controller itself
 - ▶ Limitations at both mechanical (e.g., torque limiters) and hardware (e.g., size and technology of the actuators and available electrical power) levels

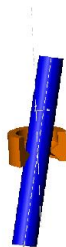
Local effect	Consequence	Action	Criticism
force applied = software threshold	force limited by the software controller	sound or light warning the surgeon	tolerable: minor trauma to human
force applied > software threshold	force applied uncontrolled by the software controller	arm stopped or automatically removed and waiting for the cause suppression	serious: injury a human
force applied > F/T sensor threshold	possible breakdown of the F/T sensor	arm stopped and power switched off	very serious: disable a human
Force applied > motor torque limiter	Motor uncontrolled	Arm stopped and power switched off	Very serious: disable a human
Force applied > mechanical resistance	Breakdown of the mechanical structure	Arm stopped and power switched off	Disastrous: decease of one or more persons in the working field

Fault tolerance

- ▶ Depending on the occurring event and its hazard, which actions may be engaged?
 - ▶ Arm deceleration until the motion stops
 - ▶ Current action cancellation
 - ▶ Placing the system in a waiting position with immobile arm until resolving the problem
 - ▶ Emergency stop with power off (stopping the arm and the hardware controllers, reinitializing variables and authorizing powering on as soon as the problem is solved)
- ▶ At each step of system design, perform a risk analysis with analytical methods
 - ▶ Failure Mode Effects and Critically Analysis (FMECA) method [GV02]. An official safety methodology used in the aeronautic and nuclear industries
 - ▶ Fault Tree Analysis (FTA)

Surgical robot design

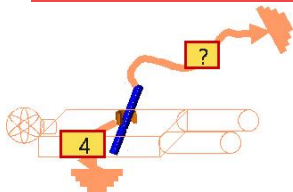
- ▶ Arms with kinematic constraints
 - ▶ The idea is to create a mechanical structure able to give the tool one specific type of motion
 - ▶ An alternative is to create artificial constraints through proper control design (e.g., stiff haptic points with artificial potential fields in the vicinity)
 - ▶ Mostly applied to mini-invasive surgery
- ▶ In MIS the tool passes through a trocar
 - ▶ The tool axis always passes in one “fixed” point
 - ▶ There are two constraints (translation is constrained in two directions)



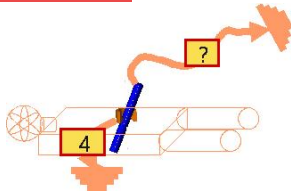
How many dof to respect the constraints?

- ▶ Chebychev–Grübler–Kutzbach criterion also called the mobility formula solves this problem [MS10]

$$\text{Mobility} = (\text{Total dof}) - (6 \times \text{Nb of loops})$$



- ▶ Mobility = **3** for T_x , T_y and T_z
- ▶ **3** = $(4 + ?) - (6 \times 1)$
- ▶ $? = 5$

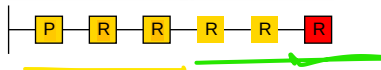


- ▶ Mobility = **4** for T_x , T_y , T_z and R_z
- ▶ **4** = $(4 + ?) - (6 \times 1)$
- ▶ $? = 6$

Option 1: passive joints

Zeus surgical robot

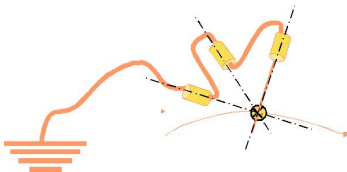
- ▶ Mobility = **3** for T_x , T_y and T_z
- ▶ **3** = $(4 + \mathbf{5}) - (6 \times 1)$
- ▶ **5** joints - **3** motors = **2** passive joints
- ▶ The first three joints are active (PRR), followed by two passive joints (RR)
- ▶ There is an optional active joint at the extremity for R_z (in this case, mobility = **4**)



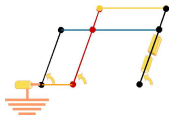
Option 2: remote rotation center (RCC)



- ▶ A classical spherical wrist does not rotate at the “right” point



- ▶ A RCC system does and thus “cancel” the constraint



- ▶ RCC with spherical links requires complex parts, while a basic parallelogram may do the job as well (e.g., Da Vinci system)
- ▶ Mobility = 6 for T_x , T_y , T_z , R_x , R_y and R_z
- ▶ $6 = (4 + 8) - (6 \times 1)$

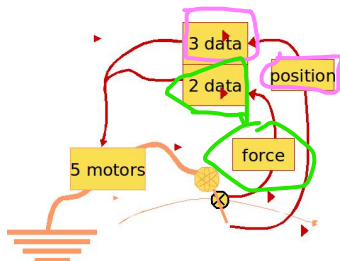


6 degrees + 2 rotation

Option 3: force control

20 for
sensor

- ▶ Mobility = **3** for T_x , T_y and T_z
- ▶ **3** = $(4 + \mathbf{5}) - (6 \times 1)$
- ▶ **5** joints and **5** motors
- ▶ Two dof controlled with force measurements and three dof controlled with position data
- ▶ All 5 motors are explicitly or implicitly controlled
- ▶ An extra force sensor can be added at the tool (distal force sensing) to measure task forces



Arms with kinematic constraints

- ▶ Option 1 (passive joints)
 - ▶ Few motors
 - ▶ The trocar “forces” the passive joint to adapt “mechanically”
 - ▶ No accurate positioning is needed
- ▶ Option 2 (RRC)
 - ▶ Few joints and motors
 - ▶ The trocar has no influence on the arm motion, but the arm must be precisely located (positioning device + procedure)
- ▶ Option 3 (force control)
 - ▶ Forces at the trocar and task variables (e.g., position and force) control all active joints – no passive joints involved
 - ▶ A bit more complex
 - ▶ May open a path to “multi-purpose” systems

Multi-purpose slave arm I

Number of dof

- ▶ Brain surgery
 - ▶ 5 dof (a point & one direction)
- ▶ Orthopedic surgery
 - ▶ 5 dof (drilling) or 6 dof (cutting)
- ▶ MIS
 - ▶ 5 dof extra-body + 3 dof (intra-body rotations)
 - ▶ 6 dof extra-body (if R_z is accounted for) + 2 dof (intra-body)
- ▶ Skin grafting - 6 dof

Speed and acceleration

- ▶ Brain surgery often works at rest
- ▶ Skin grafting and orthopedic surgery requires a few mm/s
- ▶ MIS requires several 100 mm/s (large tool rotations)
- ▶ Beating heart surgery requires accelerations greater than 1 g

Multi-purpose slave arm II

Forces

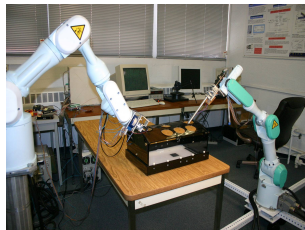
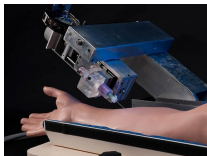
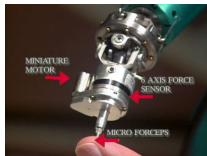
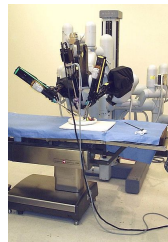
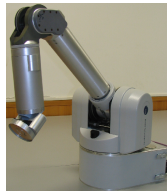
- ▶ Brain surgery requires small and medium forces
- ▶ Skin grafting requires 40 [N] to 80 [N]
- ▶ MIS requires small forces (few [N]) + disturbances due to trocar constraints
- ▶ Orthopedic surgery requires medium forces (up to 100 [N]), extremely dependent on cutting parameters

Force & speed

- ▶ Good behavior at speed as low as few [mm/s]
- ▶ Capability to move as fast as several 100 [mm/s]
- ▶ Sensitivity good enough to guarantee low forces
- ▶ Good stiffness and capability to exert forces up to 100 [N]
- ▶ Good acceleration ability

Slave arms

Lightweight, custom and industrial arms



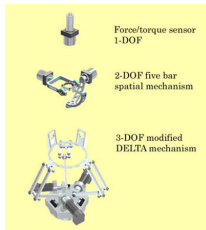
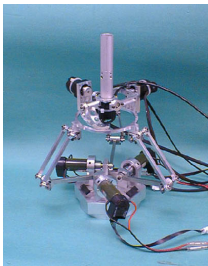
Multi-purpose master arm

General specifications

- ▶ Number of dof: 6
- ▶ Range of motion
 - ▶ Translation: $10 \times 10 \times 10$ [cm] to $20 \times 20 \times 20$ [cm]
 - ▶ Rotation: $(140 \times 90 \times 120)$ degrees
- ▶ Force-feedback
 - ▶ Several dof for force sensing (e.g., orthopedic and MIS)
 - ▶ High dynamics (e.g., touch of a finger on atheromatous arteries for heart surgery)
- ▶ High dynamics and extreme sensitivity makes it difficult for serial arms (incredible costs!)
- ▶ No “pure” parallel kinematics arm exists with such a large range of motion. Hybrid design might be a solution

Master arms

Parallel and serial arms



Chapter 3

Fundamental relations for robot control

This chapter addresses rigid robot dynamics [SC14b] in both joint and task spaces. The description of robot dynamics in joint space is a classical robotics problem, being analyzed in many textbooks (e.g. Spong and Vidyasagar [SV89], Yoshikawa [Yos90] and Sciavicco and Siciliano [SS00]). Operational space techniques introduced by Khatib [Kha83], [Kha87] are the basis for computed torque methods in the task space. Modern dynamic modeling identification methods have been presented in [SC19], [SC14b].

Matrix derivatives I

- If $A(x)$ is a matrix ($m \times n$) and x is a vector ($n \times 1$), where

$$A(x) = \begin{bmatrix} a_{11}(x) & \dots & a_{1n}(x) \\ \vdots & \ddots & \vdots \\ a_{m1}(x) & \dots & a_{mn}(x) \end{bmatrix} \quad (3.1)$$

the derivative of $A(x)$ with respect to x_i ($i = 1 \dots n$) is given by

$$\frac{dA(x)}{dx_i} = \begin{bmatrix} \frac{da_{11}(x)}{dx_i} & \dots & \frac{da_{1n}(x)}{dx_i} \\ \vdots & \ddots & \vdots \\ \frac{da_{m1}(x)}{dx_i} & \dots & \frac{da_{mn}(x)}{dx_i} \end{bmatrix} \quad (3.2)$$

with

$$x = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}^T \quad (3.3)$$

Matrix derivatives II

- ▶ The product between x^T and $\frac{dA^T(x)}{dx}$ is a matrix ($n \times m$) given by

$$x^T \frac{dA^T(x)}{dx} = \begin{bmatrix} x^T \frac{dA^T(x)}{dx_1} \\ \vdots \\ x^T \frac{dA^T(x)}{dx_n} \end{bmatrix} \quad (3.4)$$

- ▶ If $x(t)$ is a vector ($n \times 1$) and $A(t)$ is a matrix ($m \times n$), then

$$\frac{d}{dt} (A(t)x(t)) = \dot{A}(t)x(t) + A(t)\dot{x}(t) \quad (3.5)$$

- ▶ For a square and invertible matrix A , the differential of A^{-1} , dA^{-1} , is obtained by differentiating $A^{-1}A = I$, which gives

$$dA^{-1}A + A^{-1}dA = 0 \rightarrow dA^{-1} = -A^{-1}dA A^{-1} \quad (3.6)$$

Matrix derivatives III

- ▶ If $B(x)$ is a scalar and x a vector ($n \times 1$), then

$$\frac{dB(x)}{dx} = \left[\frac{dB(x)}{dx_1} \quad \cdots \quad \frac{dB(x)}{dx_n} \right]^T \quad (3.7)$$

- ▶ If $B(x)$ is a scalar and x a vector ($n \times 1$) function of time, then

$$\frac{dB(x(t))}{dt} = \dot{x}^T \frac{\partial B(x)}{\partial x} \quad (3.8)$$

- ▶ If $Y(x)$ is a vector ($m \times 1$) and x a vector ($n \times 1$),

$$\frac{dY^T(x)}{dx} = \begin{bmatrix} \frac{dY^T(x)}{dx_1} \\ \vdots \\ \frac{dY^T(x)}{dx_n} \end{bmatrix} \quad (3.9)$$

Newton's laws of motion I

1. First law (law of inertia). Every object persists in its state of rest or uniform motion in a straight line unless it is compelled to change that state by forces impressed on it
 2. Second law. Force is equal to the change in momentum (mv) per change in time. For a constant mass, force equals mass times acceleration
 3. Third law. For every action, there is an equal and opposite reaction
- ▶ Newton's laws assume an inertial reference frame
 - ▶ The first and third Newton's laws are a consequence of the second Newton's law

Newton's laws of motion II

- ▶ From the second Newton's Law, the sum of all forces F acting on a body of mass m is equal to:

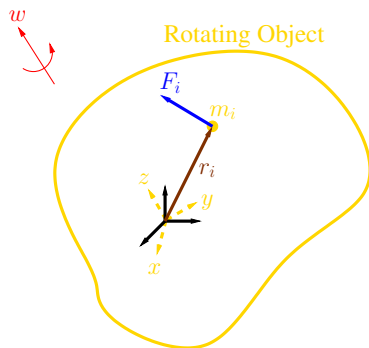
$$F = m a \quad (3.10)$$

where a is the body acceleration. If $F = 0$, we get $a = 0$, which corresponds to the first Newton's law

- ▶ Without losing generality, if we apply an external force F_e to a body of mass m_1 which is in turn attached to another body of mass m_2 , both bodies get the same acceleration $a = F_e / (m_1 + m_2)$. Therefore, m_2 receives the force $F_2 = m_2 F_e / (m_1 + m_2)$ and m_1 receives the force $F_1 = m_1 F_e / (m_1 + m_2)$. This means that m_1 applied the force F_2 on m_2 , and in turn m_2 applied the force $-F_2$ on m_1 , so that $F_e - F_2 = F_1$, which is the third Newton's law

Euler's formula I

- ▶ Inertial (black) and body attached (yellow) coordinate systems are placed at the pivot point
- ▶ The object is rotating w.r.t. ground with angular velocity w . This vector w is represented in the body attached frame, which rotates with the object
- ▶ Applying second Newton's law to a rotating body, Euler's formula is obtained, as well as the inertia tensor concept



Euler's formula II

- ▶ Let's consider the particle $m_i = \rho_i dV$, where ρ_i is the object mass density and dV is an infinitesimal volume, has a total applied force. This vector force in inertial frame coordinates is represented by 0F_i and in the body attached frame is represented by F_i (similar notation for other entities)
- ▶ Applying Newton's law,

$${}^0F_i = m_i {}^0a_i \quad (3.11)$$

where 0a_i is the acceleration w.r.t. ground represented in the inertial frame

- ▶ Relating both coordinate systems by a rotation matrix R , (3.11) can be written as

$$R^{-1} {}^0F_i = m_i R^{-1} {}^0a_i \quad (3.12)$$

which is equal to

$$F_i = m_i a_i \quad (3.13)$$

where a_i is the acceleration w.r.t. ground represented in the body attached frame

Euler's formula III

- ▶ Applying the cross-product ($r_i \times$) to (3.13), where r_i is the vector distance from the pivot point to the mass m_i , represented in the body attached frame,

$$r_i \times F_i = m_i r_i \times a_i \quad (3.14)$$

- ▶ Let's analyze now a_i . Since ${}^0v_i = {}^0w \times {}^0r_i$, where 0v_i is the velocity of m_i , 0a_i is the derivative of 0v_i ,

$${}^0a_i = {}^0\dot{w} \times {}^0r_i + {}^0w \times {}^0\dot{r}_i \quad (3.15)$$

Therefore,

$${}^0a_i = {}^0\dot{w} \times {}^0r_i + {}^0w \times ({}^0w \times {}^0r_i) \quad (3.16)$$

since ${}^0\dot{r}_i = {}^0v_i$.

Euler's formula IV

- ▶ Left-multiplying (3.16) by R^{-1} , we get

$$a_i = \dot{w} \times r_i + w \times (w \times r_i) \quad (3.17)$$

- ▶ From (3.17) and (3.14), the torque $d\tau_i = r_i \times F_i$ applied to m_i is given by

$$d\tau_i = m_i r_i \times (\dot{w} \times r_i) + m_i r_i \times [w \times (w \times r_i)] \quad (3.18)$$

- ▶ Knowing that the linear momentum $d^0 P_i = m_i {}^0 v_i$, the angular momentum $d^0 M_i = {}^0 r_i \times d^0 P_i$ is expressed as

$$d^0 M_i = m_i {}^0 r_i \times {}^0 v_i \quad (3.19)$$

Euler's formula V

- ▶ Equation (3.19) can also be written as

$$d^0 M_i = m_i {}^0 r_i \times ({}^0 w \times {}^0 r_i) \quad (3.20)$$

- ▶ Multiplying (3.20) on the left by R^{-1} , we get

$$dM_i = m_i r_i \times (w \times r_i) \quad (3.21)$$

- ▶ Making analogy with the linear case, we can define the relation between the angular momentum and the angular velocity as the moment of inertia (also called inertia tensor) dl_i , i.e.,

$$dM_i = dl_i w \quad (3.22)$$

Euler's formula VI

- From (3.22), (3.21) and (3.18),

$$d\tau_i = dl_i \dot{w} - m_i r_i \times [(w \times r_i) \times w] \quad (3.23)$$

- In general, the associative property of the cross product is not commutative, but in this case, fortunately it is. Equation (3.23) can also be written as

$$d\tau_i = dl_i \dot{w} - [m_i r_i \times (w \times r_i)] \times w \quad (3.24)$$

- From (3.21) and (3.22), (3.24) becomes

$$d\tau_i = dl_i \dot{w} - (dl_i w) \times w \quad (3.25)$$

which is equal to

$$d\tau_i = dl_i \dot{w} + w \times (dl_i w) \quad (3.26)$$

Euler's formula VII

- ▶ Integrating (3.26) over the entire body volume, and keeping in mind that the whole body rotates with the same angular velocity, we get

$$\int_V d\tau_i = \int_V dl_i \dot{w} + \int_V w \times (dl_i w) \quad (3.27)$$

which is equal to

$$\tau = I\dot{w} + w \times (Iw) \quad (3.28)$$

Equation (3.28) is known as Euler's formula

- ▶ The quantity I is the inertia tensor of the body referred to the pivot point and written in object frame coordinates

Euler's formula VIII

- Let's analyze now the inertia tensor I . From (3.21) and (3.22),

$$\int_V dl_i w = \int_V \rho_i r_i \times (w \times r_i) dV \quad (3.29)$$

which is equal to

$$Iw = \int_V \rho_i r_i \times (w \times r_i) dV \quad (3.30)$$

- Since

$$w \times r_i = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ w_x & w_y & w_z \\ r_x & r_y & r_z \end{vmatrix} \quad (3.31)$$

which gives

$$w \times r_i = \hat{x}(w_y r_x - w_z r_y) + \hat{y}(w_z r_x - w_x r_z) + \hat{z}(w_x r_y - w_y r_x) \quad (3.32)$$

Euler's formula IX

► Additionally,

$$\rho_i r_i \times (w \times r_i) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \rho_i r_x & \rho_i r_y & \rho_i r_z \\ w_y r_z - w_x r_y & w_z r_x - w_x r_z & w_x r_y - w_y r_x \end{vmatrix} \quad (3.33)$$

which gives

$$\begin{aligned} \rho_i r_i \times (w \times r_i) = & \hat{x} \rho_i \left[(r_y^2 + r_z^2) w_x - r_x r_y w_y - r_x r_z w_z \right] \\ & + \hat{y} \rho_i \left[-r_x r_y w_x + (r_x^2 + r_z^2) w_y - r_y r_z w_z \right] \\ & + \hat{z} \rho_i \left[-r_x r_z w_x - r_y r_z w_y + (r_x^2 + r_y^2) w_z \right] \end{aligned} \quad (3.34)$$

Euler's formula X

- From (3.30) and (3.34),

$$Iw = \begin{bmatrix} \int_V (r_y^2 + r_z^2) \rho_i dV & -\int_V r_x r_y \rho_i dV & -\int_V r_x r_z \rho_i dV \\ -\int_V r_x r_y \rho_i dV & \int_V (r_x^2 + r_z^2) \rho_i dV & -\int_V r_y r_z \rho_i dV \\ -\int_V r_x r_z \rho_i dV & -\int_V r_y r_z \rho_i dV & \int_V (r_x^2 + r_y^2) \rho_i dV \end{bmatrix} \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \quad (3.35)$$

- Therefore, the inertia tensor I in object coordinates referred to the pivot point is a constant and symmetric matrix given by

$$I = \begin{bmatrix} \int_V (r_y^2 + r_z^2) \rho_i dV & -\int_V r_x r_y \rho_i dV & -\int_V r_x r_z \rho_i dV \\ -\int_V r_x r_y \rho_i dV & \int_V (r_x^2 + r_z^2) \rho_i dV & -\int_V r_y r_z \rho_i dV \\ -\int_V r_x r_z \rho_i dV & -\int_V r_y r_z \rho_i dV & \int_V (r_x^2 + r_y^2) \rho_i dV \end{bmatrix} \quad (3.36)$$

Euler's formula XI

- ▶ Euler's formula can also be written in the inertial frame. Left-multiplying (3.28) by the rotation matrix R we get

$$R\tau = RI\dot{w} + R w \times (R w) \quad (3.37)$$

- ▶ Since $\dot{w} = R^{-1} {}^0\dot{w}$ and $w = R^{-1} {}^0w$,

$${}^0\tau = RIR^{-1} {}^0\dot{w} + {}^0w \times (RIR^{-1} {}^0w) \quad (3.38)$$

which can be written as

$${}^0\tau = {}^0I {}^0\dot{w} + {}^0w \times ({}^0I {}^0w) \quad (3.39)$$

where

$${}^0I = RIR^{-1} \quad (3.40)$$

- ▶ In this case, 0I is the inertia tensor of the body referred to the pivot point and written in inertial frame coordinates. 0I is not constant, since R changes as the object rotates

Euler's formula XII

- ▶ It can be inferred from this analysis that the angular motion of a rigid object can be described by a rotation around a pivot point. This pivot point might be located in, on, or outside the object
- ▶ Two reference frames are placed at this pivot point. An inertial frame and a frame that is attached to the object (i.e., it rotates with the object)
- ▶ Rotation dynamics is based on Newton's Laws. The Euler's formula in the body attached frame (3.28) or in the inertial frame (3.39) describes angular motion dynamics based on torque, angular velocity, angular acceleration, and the inertia tensor

Kinetic energy of a particle I

- ▶ Kinetic and potential energies are associated with the concept of mechanical work
- ▶ The mechanical work W is defined as

$$W = \int_{x_0}^x F^T dx \quad (3.41)$$

where x is a spacial position, F is a force vector acting on the particle and dx is the infinitesimal particle displacement vector

- ▶ Then, using the second Newton's law,

$$W = \int_{x_0}^x m \frac{dv^T}{dt} dx = \int_{x_0}^x m dv^T v = \int_{x_0}^x mv^T dv \quad (3.42)$$

- ▶ Therefore, W can be represented as

$$W = \int_{x_0}^x mv^T dv = \frac{1}{2}mv^T(x)v(x) - \frac{1}{2}mv^T(x_0)v(x_0) \quad (3.43)$$

Kinetic energy of a particle II

- ▶ The kinetic energy E at the position x is defined as

$$E(x) = \frac{1}{2}mv^T(x)v(x) \quad (3.44)$$

- ▶ Hence, the work done on a particle m is the difference between the final and initial kinetic energies

Potential energy of a particle

- ▶ The potential energy U is defined as the "negative work", i.e.,

$$U = - \int_{x_0}^x F^T dx \quad (3.45)$$

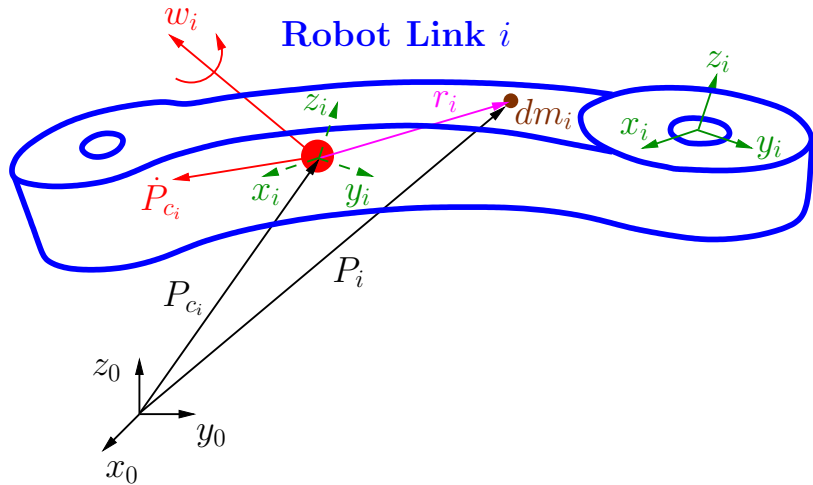
- ▶ Therefore, a positive change in the kinetic energy corresponds to a negative change in the potential energy. Also, a negative change in the kinetic energy corresponds to a positive change in the potential energy
- ▶ For instance, the work done to bring a particle m from a vertical position z_0 to another vertical position z (which is above z_0), in the presence of gravity is

$$W = \int_{z_0}^z (-mg) dz = -mg(z - z_0) < 0 \quad (3.46)$$

since the displacement vector and the gravity force point in opposite directions. The potential energy is given by

$$U = - \int_{z_0}^z (-mg) dz = mg(z - z_0) > 0 \quad (3.47)$$

Kinetic energy of a robot manipulator I



- General robot link i . P_{c_i} points to the center of mass (red dot). P_i and r_i point to a general particle of link i with infinitesimal mass dm_i . Inertial (black) and body attached coordinate systems (green) are used for analysis

Kinetic energy of a robot manipulator II

- ▶ The kinetic energy dE_i of the infinitesimal particle dm_i is given by

$$dE_i = \frac{1}{2} dm_i \dot{P}_i^T \dot{P}_i \quad (3.48)$$

which can be written as

$$dE_i = \frac{1}{2} \dot{P}_i^T \dot{P}_i \rho_i dV \quad (3.49)$$

where ρ_i is the volumetric mass density

- ▶ The velocity of dm_i , $\dot{P}_i$ is given by

$$\dot{P}_i = \dot{P}_{c_i} + w_i \times r_i = \dot{P}_{c_i} + S(w_i) r_i \quad (3.50)$$

where $\dot{P}_{c_i}$ is the linear velocity of the center of mass (and of link i) and w_i is the angular velocity of link i where the pivot point is at the center of mass. All these velocities are w.r.t. ground, although the vector representation can be done in any reference frame. $S(w_i)$ is the skew-symmetric operator

Kinetic energy of a robot manipulator III

- Therefore, the total kinetic energy of link i , E_i , is given by

$$E_i = \frac{1}{2} \int_{V_i} \left(\dot{P}_{c_i} + S(w_i)r_i \right)^T \left(\dot{P}_{c_i} + S(w_i)r_i \right) \rho_i dV \quad (3.51)$$

- Manipulating (3.51) we get

$$\begin{aligned} E_i = & \frac{1}{2} m_i \dot{P}_{c_i}^T \dot{P}_{c_i} + 2 \frac{1}{2} \dot{P}_{c_i}^T S(w_i) \int_{V_i} (P_i - P_{c_i}) \rho_i dV \\ & + \frac{1}{2} \int_{V_i} r_i^T S(w_i)^T S(w_i) r_i \rho_i dV \end{aligned} \quad (3.52)$$

Since

$$\frac{1}{m_i} \int_{V_i} P_i \rho_i dV = P_{c_i} \quad (3.53)$$

where m_i is the total mass of link i , the second term of (3.52) is zero. Hence,

$$E_i = \frac{1}{2} m_i \dot{P}_{c_i}^T \dot{P}_{c_i} + \frac{1}{2} \int_{V_i} r_i^T S(w_i)^T S(w_i) r_i \rho_i dV \quad (3.54)$$

Kinetic energy of a robot manipulator IV

- Knowing that $S(w_i)r_i = -S(r_i)w_i$, (3.54) can also be written as

$$E_i = \frac{1}{2}m_i\dot{P}_{c_i}^T\dot{P}_{c_i} + \frac{1}{2}\int_{V_i} w_i^T S(r_i)^T S(r_i)w_i \rho_i dV \quad (3.55)$$

Since the whole link i (rigid body) rotates with the same angular velocity, (3.55) can be written as

$$E_i = \frac{1}{2}m_i\dot{P}_{c_i}^T\dot{P}_{c_i} + \frac{1}{2}w_i^T \int_{V_i} S(r_i)^T S(r_i)\rho_i dV w_i \quad (3.56)$$

or simply

$$E_i = \frac{1}{2}m_i\dot{P}_{c_i}^T\dot{P}_{c_i} + \frac{1}{2}w_i^T I_i w_i \quad (3.57)$$

where I_i is the inertia tensor relative to the center of mass (pivot point)

Kinetic energy of a robot manipulator V

- In fact, since

$$S(r_i) = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix} \quad (3.58)$$

I_i represented by

$$I_i = \int_{V_i} S(r_i)^T S(r_i) \rho_i dV \quad (3.59)$$

becomes

$$I_i = \begin{bmatrix} \int_{V_i} (r_y^2 + r_z^2) \rho_i dV & -\int_{V_i} r_x r_y \rho_i dV & -\int_{V_i} r_x r_z \rho_i dV \\ -\int_{V_i} r_x r_y \rho_i dV & \int_{V_i} (r_x^2 + r_z^2) \rho_i dV & -\int_{V_i} r_y r_z \rho_i dV \\ -\int_{V_i} r_x r_z \rho_i dV & -\int_{V_i} r_y r_z \rho_i dV & \int_{V_i} (r_x^2 + r_y^2) \rho_i dV \end{bmatrix} \quad (3.60)$$

which is the same result of (3.36)

Kinetic energy of a robot manipulator VI

- ▶ The kinetic energy (3.57) has two terms. One due to the linear velocity and another one due to the angular velocity. These terms can be computed in any coordinate frame
- ▶ If we compute the angular velocity in the body attached coordinate system placed at the center of mass, I_i is constant. In this case, I_i is the inertia tensor relative to the center of mass expressed in link i coordinates. In this scenario, w_i is also expressed in the body attached coordinate system
- ▶ Since, ${}^0w_i = R_i w_i$, where R_i is a rotation matrix, (3.57) can be written as

$$E_i = \frac{1}{2} m_i \dot{P}_{c_i}^T \dot{P}_{c_i} + \frac{1}{2} {}^0w_i^T R_i I_i R_i^T {}^0w_i \quad (3.61)$$

- ▶ In this case the inertia tensor relative to the center of mass expressed in the inertial frame, 0I_i , is given by

$${}^0I_i = R_i I_i R_i^T \quad (3.62)$$

0I_i is not constant, and changes as link i rotates

Kinetic energy of a robot manipulator VII

- Relating linear $\dot{P}_{c_i}$ and angular 0w_i velocities with robot joint velocities $\dot{q}$, we get

$$\dot{P}_{c_i} = J_{c_i}(q)\dot{q} \quad (3.63)$$

and

$${}^0w_i = J_{w_i}(q)\dot{q} \quad (3.64)$$

where J_{c_i} and J_{w_i} are respectively the linear and angular Jacobians referred to the center of mass of link i .

Incorporating (3.63) and (3.64) into (3.61), we get

$$E_i = \frac{1}{2}\dot{q}^T \left[m_i J_{c_i}^T J_{c_i} + J_{w_i}^T R_i I_i R_i^T J_{w_i} \right] \dot{q} \quad (3.65)$$

- For a robot with n links, the total robot kinetic energy E is therefore given by

$$E = \frac{1}{2}\dot{q}^T \left[\sum_{i=1}^n \left(m_i J_{c_i}^T J_{c_i} + J_{w_i}^T R_i I_i R_i^T J_{w_i} \right) \right] \dot{q} \quad (3.66)$$

Kinetic energy of a robot manipulator VIII

- Defining the robot inertia matrix $M(q)$ as

$$M(q) = \sum_{i=1}^n \left(m_i J_{c_i}^T J_{c_i} + J_{w_i}^T R_i I_i R_i^T J_{w_i} \right) \quad (3.67)$$

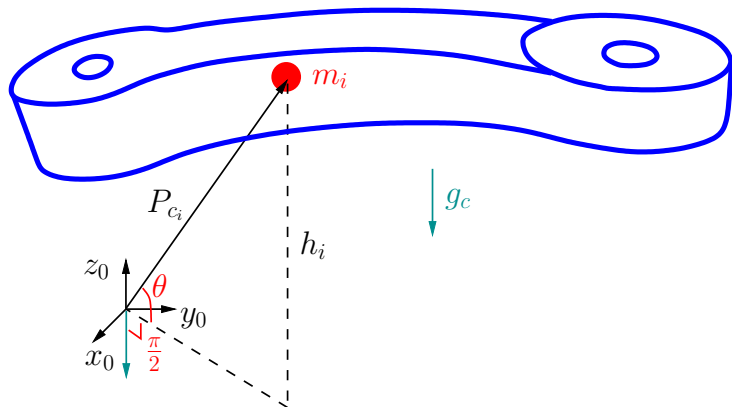
Equation (3.66) can be written as

$$E(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (3.68)$$

which is the compact representation for the robot kinetic energy

Potential energy of a robot manipulator I

Robot Link i



- The potential energy of link i depends on the height that the center of mass is from the ground level. g_c is the gravity force vector, and the inertial frame is placed at the ground, where the potential energy is conventionally defined as zero

Potential energy of a robot manipulator II

- The height of the center of mass h_i is given by

$$h_i = \|P_{c_i}\| \sin \theta = -\|P_{c_i}\| \cos(\theta + \pi/2) \quad (3.69)$$

Since the angle between P_{c_i} and g_c is $\theta + \pi/2$, the potential energy of link i , U_i can be computed as

$$U_i = -m_i P_{c_i}^T g_c = -m_i \|P_{c_i}\| \|g_c\| \cos(\theta + \pi/2) = m_i \|g_c\| h_i \quad (3.70)$$

- Therefore, the total potential energy $U(q)$ of a robot manipulator with n links is given by

$$U(q) = \sum_{i=1}^n U_i = - \sum_{i=1}^n m_i P_{c_i}^T g_c \quad (3.71)$$

Euler-Lagrange equations I

- ▶ Euler-Lagrange equations of motion are based on the principle of least action. This principle was proposed by Leibniz, and later on by Maupertuis and Euler
- ▶ The principle of least action says [Pen07]: "The path taken by the system between times t_1 and t_2 and configurations q_1 and q_2 is the one for which the action is stationary (no change) to first order." This principle assumes a natural and conservative system (where friction and other external forces are zero)
- ▶ Lagrange applied this principle to dynamic systems through the Lagrangian formulation
- ▶ Lagrange defined the action $P(q(t), \dot{q}(t), t)$ as

$$P(q, \dot{q}, t) = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad (3.72)$$

where q represent a vector of generalized coordinates, and the Lagrangian $L(q, \dot{q}, t)$ is given by

$$L(q, \dot{q}, t) = E(q, \dot{q}) - U(q) \quad (3.73)$$

Euler-Lagrange equations II

- ▶ Slightly changing (3.72) to

$$\bar{P}(q, \dot{q}, t) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad (3.74)$$

provides the insight that $\bar{P}(q, \dot{q}, t)$ is the average value of the kinetic energy minus the average value of the potential energy for the period $[t_1, t_2]$

- ▶ Let's seek the function $q_0(t)$ that minimizes (3.72) for known initial and final states q_1, t_1 and q_2, t_2 , respectively
- ▶ Let's define the function [Mor03]

$$q(t) = q_0(t) + \alpha \beta(t) \quad (3.75)$$

with

$$\beta(t_1) = \beta(t_2) = 0 \quad (3.76)$$

- ▶ The derivative of (3.72) w.r.t. α should therefore be zero, i.e.,

$$\frac{dP(q(t))}{d\alpha} = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \frac{\partial q}{\partial \alpha} + \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial \alpha} \right) dt = 0 \quad (3.77)$$

Euler-Lagrange equations III

- Straightforward manipulation of (3.77) gives

$$\frac{dP(q(t))}{d\alpha} = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \beta(t) + \frac{\partial L}{\partial \dot{q}} \dot{\beta}(t) \right) dt \quad (3.78)$$

- Integrating by parts the last term of (3.78),

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \dot{\beta}(t) dt = \left[\beta(t) \frac{\partial L}{\partial \dot{q}} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \beta(t) \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) dt \quad (3.79)$$

- From (3.76),

$$\left[\beta(t) \frac{\partial L}{\partial \dot{q}} \right]_{t_1}^{t_2} = 0 \quad (3.80)$$

Euler-Lagrange equations IV

- Therefore, from (3.78)-(3.80),

$$\frac{dP(q(t))}{d\alpha} = \int_{t_1}^{t_2} \frac{\partial L}{\partial q} \beta(t) dt - \int_{t_1}^{t_2} \beta(t) \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) dt \quad (3.81)$$

which can be written as

$$\frac{dP(q(t))}{d\alpha} = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \beta(t) dt \quad (3.82)$$

- From (3.77) and (3.82) we get the Euler-Lagrange equations of motion for a conservative system given by

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad (3.83)$$

Euler-Lagrange equations V

- ▶ Since the general form of the kinetic energy for a system represented by q generalized coordinates is given by

$$E(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (3.84)$$

where $M(q)$ is the inertia matrix, (3.83) can be written in the Newtonian form (i.e., using second Newton's Law) as

$$M(q)\ddot{q} = -\dot{M}(q)\dot{q} + \frac{\partial E(q, \dot{q})}{\partial q} - \frac{\partial U}{\partial q} \quad (3.85)$$

where the terms on the right side of (3.85) represent all natural and conservative forces present in the system

- ▶ Merging all other external forces (including friction and artificially generated forces) in the generalized vector τ , we can add τ to the right side of (3.85), entailing the general Euler-Lagrange equations given by

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = \tau \quad (3.86)$$

Manipulator dynamics in the joint space I

- ▶ Robot dynamics can be obtained by Euler-Lagrange or recursive Newton-Euler techniques
- ▶ The Lagrangian mechanics is a re-formulation of Newton's laws (classical mechanics) which takes into account energy relations
- ▶ The Lagrangian of the robot L is defined by

$$L(q, \dot{q}) = E(q, \dot{q}) - U(q) \quad (3.87)$$

where $E(q, \dot{q})$ and $U(q)$ are scalars representing respectively robot kinetic and potential energies. q is a vector ($n \times 1$) of generalized coordinates, representing the position of robot links

- ▶ The robot kinetic energy is given by (see (3.68))

$$E(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (3.88)$$

where $M(q)$ is the mass matrix ($n \times n$), which is symmetric and positive definite

Manipulator dynamics in the joint space II

- ▶ The robot potential energy is given by (see (3.71))

$$U(q) = -g_c^T r_c(q) m_c \quad (3.89)$$

where g_c is the gravity acceleration vector (3×1), $r_c(q)$ is the links' center of gravity matrix ($3 \times n$) and m_c is the links' mass vector ($n \times 1$)

- ▶ The Lagrangian mechanics is then represented in vector form by (see (3.86))

$$\tau = \frac{d}{dt} \frac{\partial L(q, \dot{q})}{\partial \dot{q}} - \frac{\partial L(q, \dot{q})}{\partial q} \quad (3.90)$$

where τ is the generalized torque acting on q

- ▶ From (3.87), (3.88) and (3.90),

$$\frac{d}{dt} (M(q)\dot{q}) - \frac{\partial E(q, \dot{q})}{\partial q} + \frac{\partial U(q)}{\partial q} = \tau \quad (3.91)$$

Manipulator dynamics in the joint space III

- ▶ Equation (3.91) can be written as

$$M(q)\ddot{q} + v(q, \dot{q}) + g(q) = \tau \quad (3.92)$$

with

$$v(q, \dot{q}) = C(q, \dot{q})\dot{q} = \dot{M}(q)\dot{q} - \frac{\partial E(q, \dot{q})}{\partial q} \quad (3.93)$$

and

$$g(q) = \frac{\partial U(q)}{\partial q} \quad (3.94)$$

The matrix $C(q, \dot{q})$ has dimensions $(n \times n)$ and is given by

$$C(q, \dot{q}) = \dot{M}(q) - \frac{1}{2}\dot{q}^T \frac{\partial M(q)}{\partial q} \quad (3.95)$$

- ▶ There are several ways to compute $C(q, \dot{q})$. The *Christoffel symbols* technique is a popular one

Mass matrix

- ▶ The mass (or inertia) matrix is given by

$$M(q) = \sum_{i=1}^n \left[m_i J_{v_i}(q)^T J_{v_i}(q) + J_{w_i}(q)^T R_i(q) I_i R_i(q)^T J_{w_i}(q) \right] \quad (3.96)$$

where m_i , $J_{v_i}(q)$, $J_{w_i}(q)$, $R_i(q)$ and I_i are respectively the mass of each link, the Jacobians associated with linear and angular velocities, the rotation matrix in base coordinates and the inertia tensor in link coordinates. All these quantities are referred to the center of mass of link i

- ▶ $M(q)$ can be explicitly computed by (3.96) or by RNE techniques. Using RNE, the first column of $M(q)$ is the value of τ for $\ddot{q} = [1 \ 0 \ 0 \ \dots \ 0]$ and $\dot{q} = 0$ subtracted by $g(q)$ (similar analysis for the other columns)

Properties of the mass matrix I

- From (3.96), the mass matrix $M(q)$ is symmetric and square ($n \times n$). Therefore

$$M(q) = M(q)^T \quad (3.97)$$

$$\left(M(q)^{-1}\right)^T = \left(M(q)^T\right)^{-1} \quad (3.98)$$

- From (3.97) and (3.98), $M(q)^{-1}$ is symmetric too
- From (3.88), $M(q)$ is positive definite since $E(q, \dot{q}) > 0$ for $\dot{q} \neq 0$
- For general vectors x ($n \times 1$) and y ($n \times 1$),

$$\frac{d}{dx} \left(x^T M(q) x \right) = 2M(q)x \quad (3.99)$$

$$\frac{d}{d\dot{x}} \left(\dot{x}^T M(q) \dot{x} \right) = 2M(q)\dot{x} \quad (3.100)$$

$$y^T M(q) x = x^T M(q) y \quad (3.101)$$

Properties of the mass matrix II

- ▶ $M(q)^{-1}$ always exists (every positive definite matrix is invertible and its inverse is also positive definite)
- ▶ The derivative of the inverse of $M(q)$ is given by (see (3.6))

$$\frac{d}{dq} \left(M(q)^{-1} \right) = -M(q)^{-1} \frac{d}{dq} (M(q)) M(q)^{-1} \quad (3.102)$$

- ▶ The matrix $N(q, \dot{q})$ defined by

$$N(q, \dot{q}) = \dot{M}(q) - 2C(q, \dot{q}) \quad (3.103)$$

is skew symmetric, i.e.,

$$N(q, \dot{q}) + N(q, \dot{q})^T = 0 \quad (3.104)$$

where $C(q, \dot{q})$ is given by (3.95). This property is important for stability analysis

Jacobian I

- ▶ The Jacobian $J(q)$ maps joint differential displacements (dq) into task differential displacements (dX). The task coordinates represent robot tasks in a more natural way (e.g., Cartesian coordinates of the end-effector), while joint coordinates represent joint displacements (e.g., angular displacements for revolute joints). $J(q)$ is represented by

$$dX = J(q)dq \quad (3.105)$$

- ▶ Equation (3.105) can also be written as

$$\dot{X} = J(q)\dot{q} \quad (3.106)$$

- ▶ The Jacobian is useful for singularity identification, redundancy analysis, force/torque mapping and implicit inverse kinematics algorithms

Jacobian II

- ▶ If q is a vector ($n \times 1$) and X is a vector ($m \times 1$), then

$$J(q) = \begin{bmatrix} J_{11}(q) & \dots & J_{1n}(q) \\ \vdots & \ddots & \vdots \\ J_{m1}(q) & \dots & J_{mn}(q) \end{bmatrix} = \begin{bmatrix} \frac{\partial X_1(q)}{\partial q_1} & \dots & \frac{\partial X_1(q)}{\partial q_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial X_m(q)}{\partial q_1} & \dots & \frac{\partial X_m(q)}{\partial q_n} \end{bmatrix} \quad (3.107)$$

- ▶ The time derivative of the Jacobian $\dot{J}(q)$ is given by

$$\dot{J}(q) = \begin{bmatrix} \dot{J}_{11}(q) & \dots & \dot{J}_{1n}(q) \\ \vdots & \ddots & \vdots \\ \dot{J}_{m1}(q) & \dots & \dot{J}_{mn}(q) \end{bmatrix} \quad (3.108)$$

where

$$\dot{J}_{ij} = \dot{q}^T \frac{\partial J_{ij}(q)}{\partial q} \quad (3.109)$$

for $i = 1 \dots m$ and $j = 1 \dots n$

Jacobian III

- ▶ If X and q have the same dimensions ($n \times 1$), the Jacobian inverse might exist, and is given by

$$dq = J^{-1}(X)dX \quad (3.110)$$

- ▶ Equation (3.110) can also be written as

$$\dot{q} = J^{-1}(X)\dot{X} \quad (3.111)$$

with

$$J^{-1}(X) = \begin{bmatrix} J_{11}^{-1}(X) & \cdots & J_{1n}^{-1}(X) \\ \vdots & \ddots & \vdots \\ J_{n1}^{-1}(X) & \cdots & J_{nn}^{-1}(X) \end{bmatrix} = \begin{bmatrix} \frac{\partial q_1(X)}{\partial X_1} & \cdots & \frac{\partial q_1(X)}{\partial X_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial q_n(X)}{\partial X_1} & \cdots & \frac{\partial q_n(X)}{\partial X_n} \end{bmatrix} \quad (3.112)$$

Jacobian IV

- The time derivative of $J^{-T}(X)$, $\dot{J}^{-T}(X)$, is given by

$$\dot{J}^{-T}(X) = \dot{X}^T \frac{\partial J^{-T}}{\partial X} = \begin{bmatrix} \dot{X}^T \frac{\partial J^{-T}}{\partial X_1} \\ \vdots \\ \dot{X}^T \frac{\partial J^{-T}}{\partial X_n} \end{bmatrix} \quad (3.113)$$

- To prove (3.113), let's analyze the i -th row of $\dot{J}^{-T}(X)$, $\dot{J}_i^{-T}(X)$. The i -th row of $J^{-T}(X)$, $J_i^{-T}(X)$, is given by (see (3.112))

$$J_i^{-T}(X) = \begin{bmatrix} \frac{\partial q_1(X)}{\partial X_i} & \dots & \frac{\partial q_n(X)}{\partial X_i} \end{bmatrix} \quad (3.114)$$

Therefore, differentiating (3.114) with respect to time gives

$$\dot{J}_i^{-T}(X) = \begin{bmatrix} \frac{\partial}{\partial X_1} \frac{\partial q_1(X)}{\partial X_i} \dot{X}_1 + \dots + \frac{\partial}{\partial X_n} \frac{\partial q_1(X)}{\partial X_i} \dot{X}_n \\ \vdots \\ \frac{\partial}{\partial X_1} \frac{\partial q_n(X)}{\partial X_i} \dot{X}_1 + \dots + \frac{\partial}{\partial X_n} \frac{\partial q_n(X)}{\partial X_i} \dot{X}_n \end{bmatrix}^T \quad (3.115)$$

Jacobian V

- On the other hand,

$$\frac{\partial J^{-T}(X)}{\partial X_i} = \begin{bmatrix} \frac{\partial}{\partial X_i} \frac{\partial q_1(X)}{\partial X_1} & \cdots & \frac{\partial}{\partial X_i} \frac{\partial q_1(X)}{\partial X_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial X_i} \frac{\partial q_n(X)}{\partial X_1} & \cdots & \frac{\partial}{\partial X_i} \frac{\partial q_n(X)}{\partial X_n} \end{bmatrix}^T \quad (3.116)$$

Multiplying (3.116) by $\dot{X}^T$,

$$\dot{X}^T \frac{\partial J^{-T}(X)}{\partial X_i} = \begin{bmatrix} \frac{\partial}{\partial X_i} \frac{\partial q_1(X)}{\partial X_1} \dot{X}_1 + \cdots + \frac{\partial}{\partial X_i} \frac{\partial q_1(X)}{\partial X_n} \dot{X}_n \\ \vdots \\ \frac{\partial}{\partial X_i} \frac{\partial q_n(X)}{\partial X_1} \dot{X}_1 + \cdots + \frac{\partial}{\partial X_i} \frac{\partial q_n(X)}{\partial X_n} \dot{X}_n \end{bmatrix}^T \quad (3.117)$$

Jacobian VI

- ▶ Equations (3.117) and (3.115) are equal, therefore (3.113) is proved. In the proof it was assumed that

$$\frac{\partial^2}{\partial X_i \partial X_j} = \frac{\partial^2}{\partial X_j \partial X_i} \quad (3.118)$$

for $i, j = 1 \dots n$

- ▶ The relation (3.118) holds if all mixed second order partial derivatives are continuous (Clairaut's theorem)
- ▶ Another useful relation is

$$\dot{J}^{-1} J = -J^{-1} \dot{J} \quad (3.119)$$

which can be obtained by differentiating $J^{-1} J = I$ with respect to time

Manipulator dynamics in the task space for non-redundant robots I

- ▶ The Lagrangian of the robot L can also be defined using the task coordinates X , if they are able to describe the whole system.

$$L(X, \dot{X}) = E(X, \dot{X}) - U(X) \quad (3.120)$$

where $E(X, \dot{X})$ and $U(X)$ are scalars representing respectively robot kinetic and potential energies. X is the task space vector ($n \times 1$), which is related to the generalized joint coordinates q by the function f ,

$$X = f(q) \quad \rightarrow \quad q = f^{-1}(X) \quad (3.121)$$

- ▶ The robot kinetic energy is given by

$$E(X, \dot{X}) = \frac{1}{2} \dot{X}^T \Lambda(X) \dot{X} \quad (3.122)$$

where $\Lambda(X)$ is the mass matrix of the task space ($n \times n$), which is symmetric and positive definite

Manipulator dynamics in the task space for non-redundant robots II

- ▶ The robot kinetic energy does not depend on the generalized coordinate system. Hence, from (3.88) and (3.122),

$$\frac{1}{2} \dot{X}^T \Lambda(X) \dot{X} = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (3.123)$$

From (3.106), (3.123) becomes

$$M(q) = J(q)^T \Lambda(X) J(q) \rightarrow \Lambda(X) = J^{-T}(X) M(q) J^{-1}(X) \quad (3.124)$$

- ▶ The Lagrangian mechanics is represented in vector form by

$$F = \frac{d}{dt} \frac{\partial L(X, \dot{X})}{\partial \dot{X}} - \frac{\partial L(X, \dot{X})}{\partial X} \quad (3.125)$$

where F is the generalized force acting on X .

- ▶ From (3.120), (3.122) and (3.125),

$$\frac{d}{dt} (\Lambda(X) \dot{X}) - \frac{\partial E(X, \dot{X})}{\partial X} + \frac{\partial U(X)}{\partial X} = F \quad (3.126)$$

Manipulator dynamics in the task space for non-redundant robots III

- ▶ In the sequel, the terms of (3.126) will be analyzed in detail and compared to (3.92)
- ▶ The robot potential energy does not depend on the generalized coordinate system. Hence,

$$U(X) = U(q) \quad (3.127)$$

Differentiating (3.127) with respect to X ,

$$\begin{bmatrix} \frac{\partial U(X)}{\partial X_1} \\ \vdots \\ \frac{\partial U(X)}{\partial X_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial U(q)}{\partial q_1} \frac{\partial q_1}{\partial X_1} + \dots + \frac{\partial U(q)}{\partial q_n} \frac{\partial q_n}{\partial X_1} \\ \vdots \\ \frac{\partial U(q)}{\partial q_1} \frac{\partial q_1}{\partial X_n} + \dots + \frac{\partial U(q)}{\partial q_n} \frac{\partial q_n}{\partial X_n} \end{bmatrix} \quad (3.128)$$

From (3.112), (3.128) can be written as

$$\frac{\partial U(X)}{\partial X} = J^{-T}(X) \frac{\partial U(q)}{\partial q} \quad (3.129)$$

Manipulator dynamics in the task space for non-redundant robots IV

- Let's analyze now the term $\frac{\partial E(X, \dot{X})}{\partial X}$. From (3.122),

$$\frac{\partial E(X, \dot{X})}{\partial X} = \frac{1}{2} \dot{X}^T \frac{\partial \Lambda(X)}{\partial X} \dot{X} \quad (3.130)$$

Knowing (3.124), and simplifying Jacobian notations,

$$\frac{\partial \Lambda(X)}{\partial X_i} = \frac{\partial J^{-T}}{\partial X_i} M(q) J^{-1} + J^{-T} \frac{\partial M(q)}{\partial X_i} J^{-1} + J^{-T} M(q) \frac{\partial J^{-1}}{\partial X_i} \quad (3.131)$$

Therefore, from (3.130) and (3.131),

$$\begin{aligned} \frac{\partial E(X, \dot{X})}{\partial X_i} &= \frac{1}{2} \dot{X}^T J^{-T} \frac{\partial M(q)}{\partial X_i} J^{-1} \dot{X} + \frac{1}{2} \dot{X}^T J^{-T} M(q) \frac{\partial J^{-1}}{\partial X_i} \dot{X} \\ &\quad + \frac{1}{2} \dot{X}^T \frac{\partial J^{-T}}{\partial X_i} M(q) J^{-1} \dot{X} \end{aligned} \quad (3.132)$$

Manipulator dynamics in the task space for non-redundant robots V

- ▶ Since the transpose of a scalar is equal to the scalar, (3.132) can be written as (see also (3.111) and (3.97))

$$\frac{\partial E(X, \dot{X})}{\partial X_i} = \frac{1}{2} \dot{q}^T \frac{\partial M(q)}{\partial X_i} \dot{q} + \dot{X}^T \frac{\partial J^{-T}}{\partial X_i} M(q) \dot{q} \quad (3.133)$$

- ▶ The term $\frac{\partial M(q)}{\partial X_i}$ can be written as

$$\frac{\partial M(q)}{\partial X_i} = \frac{\partial q_1}{\partial X_i} \frac{\partial M(q)}{\partial q_1} + \dots + \frac{\partial q_n}{\partial X_i} \frac{\partial M(q)}{\partial q_n} \quad (3.134)$$

Hence,

$$\dot{q}^T \frac{\partial M(q)}{\partial X_i} \dot{q} = \left[\frac{\partial q_1}{\partial X_i} \quad \dots \quad \frac{\partial q_n}{\partial X_i} \right] \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} \quad (3.135)$$

Manipulator dynamics in the task space for non-redundant robots VI

- From (3.113), (3.135), (3.133) and (3.112),

$$\frac{\partial E(X, \dot{X})}{\partial X} = \frac{1}{2} J^{-T} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} + J^{-T} M(q) \dot{q} \quad (3.136)$$

which can be written in compact form by

$$\frac{\partial E(X, \dot{X})}{\partial X} = \frac{1}{2} J^{-T} \dot{q}^T \frac{\partial M(q)}{\partial q} \dot{q} + J^{-T} M(q) \dot{q} \quad (3.137)$$

From (3.88),

$$\frac{\partial E(X, \dot{X})}{\partial X} = J^{-T} \frac{\partial E(q, \dot{q})}{\partial q} + J^{-T} M(q) \dot{q} \quad (3.138)$$

Manipulator dynamics in the task space for non-redundant robots VII

- Let's analyze now the term $\frac{d}{dt} (\Lambda(X)\dot{X})$,

$$\frac{d}{dt} (\Lambda(X)\dot{X}) = \Lambda(X)\ddot{X} + \dot{\Lambda}(X)\dot{X} \quad (3.139)$$

Knowing (3.124),

$$\dot{\Lambda}(X)\dot{X} = \dot{J}^{-T} M(q)\dot{q} + J^{-T} \dot{M}(q)\dot{q} + J^{-T} M(q)\dot{J}^{-1} J\dot{q} \quad (3.140)$$

- From (3.119) and (3.140), (3.139) can be written as

$$\frac{d}{dt} (\Lambda(X)\dot{X}) = \Lambda(X)\ddot{X} + \dot{J}^{-T} M(q)\dot{q} + J^{-T} \dot{M}(q)\dot{q} - \Lambda(X)\dot{J}\dot{q} \quad (3.141)$$

Manipulator dynamics in the task space for non-redundant robots VIII

- ▶ Substituting (3.129), (3.138) and (3.141) in (3.126), and doing straightforward algebra gives

$$M(q)\ddot{q} + \dot{M}(q)\dot{q} - \frac{\partial E(q, \dot{q})}{\partial q} + \frac{\partial U(q)}{\partial q} = J^T F \quad (3.142)$$

- ▶ Comparing (3.142) with (3.91), the fundamental result

$$\tau = J^T F \quad (3.143)$$

is obtained, which is the basis for dynamic control

Manipulator dynamics in the task space for redundant robots I

- ▶ If X and q do not have the same dimensions, i.e., ($m < n$), the Jacobian is not a square matrix having no inverse
- ▶ Multiple $\dot{q}$ solutions exist that satisfy (3.106), for a given $\dot{X}$ and q
- ▶ Let's seek $\dot{q}$ solutions that minimize the cost function $B(\dot{q})$,

$$B(\dot{q}) = \frac{1}{2}(\dot{q} - \dot{q}_0)^T W(\dot{q} - \dot{q}_0) \quad (3.144)$$

where W is a symmetric positive definite matrix with dimensions $(n \times n)$ and $\dot{q}$ should be as close to $\dot{q}_0$ as possible

- ▶ This problem can be solved with the method of Lagrange multipliers. Consider the modified cost function $B(\dot{q}, \lambda)$,

$$B(\dot{q}, \lambda) = \frac{1}{2}(\dot{q} - \dot{q}_0)^T W(\dot{q} - \dot{q}_0) + \lambda^T (\dot{X} - J\dot{q}) \quad (3.145)$$

where λ is an $(m \times 1)$ vector of unknown multipliers that allows to incorporate the constraint (3.106)

Manipulator dynamics in the task space for redundant robots II

- ▶ Since $B(\dot{q}, \lambda)$ is a scalar,

$$B(\dot{q}, \lambda) = B(\dot{q}, \lambda)^T = \frac{1}{2}(\dot{q} - \dot{q}_0)^T W(\dot{q} - \dot{q}_0) + (\dot{X} - J\dot{q})^T \lambda \quad (3.146)$$

- ▶ The requested solution has to satisfy

$$\frac{\partial B(\dot{q}, \lambda)}{\partial \dot{q}} = 0 \quad (3.147)$$

$$\frac{\partial B(\dot{q}, \lambda)}{\partial \lambda} = 0 \quad (3.148)$$

- ▶ From (3.146) and (3.147),

$$\dot{q} = \dot{q}_0 + W^{-1} J^T \lambda \quad (3.149)$$

- ▶ From (3.146) and (3.148), the constraint is recovered

$$\dot{X} = J\dot{q} \quad (3.150)$$

Manipulator dynamics in the task space for redundant robots III

- ▶ The solution (3.149) is a minimum, since

$$\frac{\partial^2 B(\dot{q}, \lambda)}{\partial \dot{q}^2} = W \quad (3.151)$$

and W is positive definite

- ▶ From (3.149) and (3.150),

$$\dot{X} = J\dot{q}_0 + JW^{-1}J^T\lambda \quad (3.152)$$

- ▶ Solving (3.152) for λ entails

$$\lambda = \left(JW^{-1}J^T\right)^{-1}(\dot{X} - J\dot{q}_0) \quad (3.153)$$

- ▶ From (3.149) and (3.153) we get the solution

$$\dot{q} = \dot{q}_0 + W^{-1}J^T \left(JW^{-1}J^T\right)^{-1}(\dot{X} - J\dot{q}_0) \quad (3.154)$$

Manipulator dynamics in the task space for redundant robots IV

- ▶ Defining the generalized inverse of J , $J^\#$, as

$$J^\# = W^{-1} J^T (J W^{-1} J^T)^{-1} \quad (3.155)$$

(3.154) can be written as

$$\dot{q} = J^\# \dot{X} + (I_{n \times n} - J^\# J) \dot{q}_0 \quad (3.156)$$

- ▶ Note that $J^\#$ is a right generalized inverse of J , i.e.,

$$J J^\# = I_{m \times m} \quad (3.157)$$

- ▶ Multiplying (3.156) by J and knowing (3.157) the constraint (3.150) is always satisfied. Therefore, the objective specified through $\dot{q}_0$ is a secondary objective that is not necessarily reached (i.e., $\dot{q}$ might not be equal to $\dot{q}_0$)

Manipulator dynamics in the task space for redundant robots V

- ▶ If $W = I$, $J^\#$ is called the Moore–Penrose pseudoinverse or just pseudoinverse. If

$$W = M(q) \quad (3.158)$$

$J^\#$ is called the dynamically consistent generalized inverse, where $\dot{q}$ solutions minimize robot kinetic energy around $\dot{q}_0$ (see (3.144))

- ▶ The matrix

$$I - J^\# J \quad (3.159)$$

projects the vector $\dot{q}_0$ into the null space of J , enabling to generate internal motions specified by

$$(I - J^\# J)\dot{q}_0 \quad (3.160)$$

without changing end-effector behavior

Manipulator dynamics in the task space for redundant robots VI

- ▶ $\dot{q}_0$ can be specified to take advantage of redundancy

$$\dot{q}_0 = K_0 \frac{\partial \gamma(q)}{\partial q} \quad (3.161)$$

where K_0 is a positive definite matrix (or just a positive scalar) and $\gamma(q)$ is a scalar criterion function where undesired q configurations correspond to minimums, and desired q configurations (if any) correspond to maximums. $\gamma(q)$ is to be maximized by the null space task

- ▶ Since the gradient vector points away from minimums and in the direction of maximums, (3.161) points away from undesired minimums of $\gamma(q)$
- ▶ A typical objective function is the manipulability measure,

$$\gamma(q) = \sqrt{\det(JJ^T)} \quad (3.162)$$

which vanishes at singular configurations. Therefore, the gradient of (3.162) points away from singularities

Manipulator dynamics in the task space for redundant robots VII

- From (3.161) and (3.156),

$$\dot{q} = J^\# \dot{X} + (I - J^\# J) K_0 \frac{\partial \gamma(q)}{\partial q} \quad (3.163)$$

- Since

$$\dot{\gamma}(q) = \left(\frac{\partial \gamma(q)}{\partial q} \right)^T \dot{q} \quad (3.164)$$

we get

$$\dot{\gamma}(q) = \left(\frac{\partial \gamma(q)}{\partial q} \right)^T J^\# \dot{X} + \left(\frac{\partial \gamma(q)}{\partial q} \right)^T (I - J^\# J) K_0 \frac{\partial \gamma(q)}{\partial q} \quad (3.165)$$

- The last term of (3.165) should always be non-negative, contributing to increase the criterion function $\gamma(q)$ to keep it away from minimums

Manipulator dynamics in the task space for redundant robots VIII

- ▶ Let's analyze the case where $J^\#$ is the dynamically consistent generalized inverse
- ▶ The total robot energy is always equal or greater than the energy associated with the task. Therefore,

$$\frac{1}{2} \dot{q}^T M(q) \dot{q} \geq \frac{1}{2} \dot{X}^T \Lambda(q) \dot{X} \quad (3.166)$$

where the task space inertia $\Lambda(q) = \left(J M(q)^{-1} J^T \right)^{-1}$ is explained in the sequel (see (3.200)). This means that

$$\dot{q}^T \left(M - J^T \Lambda J \right) \dot{q} \geq 0 \quad (3.167)$$

- ▶ From (3.167), the matrix

$$\left(M - J^T \Lambda J \right) \succeq 0 \quad (3.168)$$

is non-negative definite

Manipulator dynamics in the task space for redundant robots IX

- ▶ Equation (3.168) can be written as (see (3.155) for $W = M$)

$$M \left(I - J^\# J \right) \succeq 0 \quad (3.169)$$

- ▶ Making $\dot{q} = M^{-1}y$ in (3.167), where y is an arbitrary vector, it can be concluded that

$$\left(I - J^\# J \right) M^{-1} \succeq 0 \quad (3.170)$$

is also non-negative definite

- ▶ Therefore, making $K_0 = \alpha M^{-1}$, where α is a scalar positive constant, the last term of (3.165) is always non-negative
- ▶ The special case $M = I$ corresponds to the Moore-Penrose pseudoinverse $J^\#$

Manipulator dynamics in the task space for redundant robots X

- ▶ It should be pointed out that for the dynamically consistent generalized inverse, the projection matrix

$$(I - J^\# J) \quad (3.171)$$

is diagonalizable with non-negative real eigenvalues. However, (3.171) can be non-Hermitian and indefinite

- ▶ More generally, if G and H are Hermitian, with H positive definite (in our case, $G = M(I - J^\# J)$, and $H = M^{-1}$), then HG is diagonalizable with real eigenvalues, and the signs of the eigenvalues of HG agree with those of G
- ▶ Let's show with a simple example that (3.171) can be indefinite and non-Hermitian, with

$$J = [1 \ 2], \quad M = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \quad (3.172)$$

Manipulator dynamics in the task space for redundant robots XI

- ▶ For redundant robots the general solution for $\dot{q}$ is (see (3.156))

$$\dot{q} = J^\# \dot{X} + (I_{n \times n} - J^\# J) \dot{q}_0 \quad (3.173)$$

- ▶ The joint displacement due to the task space is therefore given by

$$\dot{q}_t = J^\# \dot{X} \quad (3.174)$$

and the corresponding power associated to the task space P_t is

$$P_t = \left(J^\# J \dot{q} \right)^T \tau \quad (3.175)$$

- ▶ Therefore, from (3.175) the task torque τ_t is given by

$$\tau_t = J^T J^\#^T \tau \quad (3.176)$$

Manipulator dynamics in the task space for redundant robots XII

- ▶ From (3.176) we can infer that

$$F = J^{\#T} \tau \quad (3.177)$$

which is the general form of the task space force F , where (see (3.157))

$$J^{\#T} J^T = I_{m \times m} \quad (3.178)$$

- ▶ Let's seek the general τ solution based on Lagrange multipliers that satisfies (3.177) and is as close to τ_0 as possible. The scalar cost function $B(\tau, \lambda)$ is defined as

$$B(\tau, \lambda) = \frac{1}{2}(\tau - \tau_0)^T A(\tau - \tau_0) + (F - J^{\#T} \tau)^T \lambda \quad (3.179)$$

where A is a positive definite matrix

Manipulator dynamics in the task space for redundant robots XIII

- ▶ The requested solution has to satisfy

$$\frac{\partial B(\tau, \lambda)}{\partial \tau} = 0 \quad (3.180)$$

$$\frac{\partial B(\tau, \lambda)}{\partial \lambda} = 0 \quad (3.181)$$

- ▶ From (3.179) and (3.180),

$$\tau = \tau_0 + A^{-1} J^\# \lambda \quad (3.182)$$

- ▶ From (3.179) and (3.181), the constraint is recovered

$$F = J^\#{}^T \tau \quad (3.183)$$

Manipulator dynamics in the task space for redundant robots XIV

- ▶ The solution (3.182) is a minimum, since

$$\frac{\partial^2 B(\tau, \lambda)}{\partial \tau^2} = A \quad (3.184)$$

and A is positive definite

- ▶ From (3.182) and (3.183),

$$F = J^{\#T} \tau_0 + J^{\#T} A^{-1} J^{\#} \lambda \quad (3.185)$$

- ▶ Solving (3.185) for λ entails

$$\lambda = \left(J^{\#T} A^{-1} J^{\#} \right)^{-1} (F - J^{\#T} \tau_0) \quad (3.186)$$

- ▶ From (3.182) and (3.186) we get the solution

$$\tau = \tau_0 + A^{-1} J^{\#} \left(J^{\#T} A^{-1} J^{\#} \right)^{-1} (F - J^{\#T} \tau_0) \quad (3.187)$$

Manipulator dynamics in the task space for redundant robots XV

- Defining the matrix $J^\dagger$ as

$$J^\dagger = A^{-1} J^\# \left(J^{\#T} A^{-1} J^\# \right)^{-1} \quad (3.188)$$

Equation (3.187) can be written as

$$\tau = J^\dagger F + \left(I - J^\dagger J^{\#T} \right) \tau_0 \quad (3.189)$$

From (3.188),

$$J^{\#T} J^\dagger = I \quad (3.190)$$

Therefore, from (3.190) and (3.178),

$$J^\dagger = J^T \quad (3.191)$$

which entails

$$\tau = J^T F + \left(I - J^T J^{\#T} \right) \tau_0 \quad (3.192)$$

Manipulator dynamics in the task space for redundant robots XVI

- ▶ The general solution for τ represented in (3.192) shows that

$$\tau = \tau_t + \tau_N \quad (3.193)$$

where the task torque τ_t has the same form of the non-redundant case, and the null space torque τ_N is given by

$$\tau_N = \left(I - J^T J^{\#T} \right) \tau_0 \quad (3.194)$$

- ▶ The objective specified through τ_0 is a secondary objective that is not necessarily reached (i.e., τ might be different from τ_0), since τ_0 only shows up in the null space term

Manipulator dynamics in the task space for redundant robots XVII

- ▶ For redundant robots, the inertia matrix $\Lambda(q)$ is a function of q instead of X , since the overall robot posture affects the task space inertia. This matrix is well identified, relating $\ddot{X}$ and F when all other terms are zero. $\Lambda(q)$ is square, symmetric (its inverse is symmetric too) and positive definite
- ▶ Multiplying (3.92) by $JM(q)^{-1}$,

$$J\ddot{q} + JM(q)^{-1} [v(q, \dot{q}) + g(q)] = JM(q)^{-1}\tau \quad (3.195)$$

Since

$$\dot{X} = J\dot{q} \quad (3.196)$$

which implies

$$\ddot{X} = \dot{J}\dot{q} + J\ddot{q} \quad (3.197)$$

Equation (3.195) becomes

$$\ddot{X} - \dot{J}\dot{q} + JM(q)^{-1} [v(q, \dot{q}) + g(q)] = JM(q)^{-1}\tau \quad (3.198)$$

Manipulator dynamics in the task space for redundant robots XVIII

- ▶ Substituting (3.192) into (3.198) we get

$$\ddot{X} - \dot{J}\dot{q} + JM(q)^{-1} [v(q, \dot{q}) + g(q)] = JM(q)^{-1} J^T F \quad (3.199)$$

- ▶ The mass matrix can be identified by (3.199). Its inverse is given by

$$\Lambda(q)^{-1} = JM(q)^{-1} J^T \quad (3.200)$$

- ▶ The null space term does not affect the task dynamics, therefore $J^{\#T}$ satisfies the relation

$$JM(q)^{-1} (I - J^T J^{\#T}) \tau_0 = 0 \quad (3.201)$$

- ▶ For arbitrary τ_0 , (3.201) holds if

$$J^{\#T} = \Lambda(q) JM(q)^{-1} \rightarrow J^{\#} = M(q)^{-1} J^T \Lambda(q) \quad (3.202)$$

which corresponds to the $\dot{q}$ solution that minimizes robot kinetic energy around $\dot{q}_0$ (see (3.144), (3.155) for $W = M(q)$)

Manipulator dynamics in the task space - Summary I

- Summarizing for redundant robots (general case), joint space and task space dynamics are related by (see (3.92), (3.195)-(3.202))

$$M(q)\ddot{q} + v(q, \dot{q}) + g(q) = \tau \quad (3.203)$$

$$\Lambda(q)\ddot{X} + V_X(q, \dot{q}) + G_X(q) = F \quad (3.204)$$

where

$$\Lambda(q) = \left(JM(q)^{-1} J^T \right)^{-1} \quad (3.205)$$

$$V_X(q, \dot{q}) = \Lambda(q) JM(q)^{-1} v(q, \dot{q}) - \Lambda(q) \dot{J} \dot{q} \quad (3.206)$$

$$G_X(q) = \Lambda(q) JM(q)^{-1} g(q) \quad (3.207)$$

$$\tau = J^T F + (I - J^T J^\#{}^T) \tau_0 \quad (3.208)$$

$$J^\#{}^T = \Lambda(q) JM(q)^{-1} \quad (3.209)$$

$$J^\# = M(q)^{-1} J^T \Lambda(q) \quad (3.210)$$

Manipulator dynamics in the task space - Summary II

- For non-redundant robots, joint space and task space dynamics are related by (see (3.203)-(3.210))

$$M(q)\ddot{q} + v(q, \dot{q}) + g(q) = \tau \quad (3.211)$$

$$\Lambda(X)\ddot{X} + V_X(X, \dot{X}) + G_X(X) = F \quad (3.212)$$

where

$$\Lambda(X) = \left(JM(q)^{-1} J^T \right)^{-1} = J^{-T} M(q) J^{-1} \quad (3.213)$$

$$V_X(X, \dot{X}) = J^{-T} v(q, \dot{q}) - \Lambda(X) \dot{J} \dot{q} \quad (3.214)$$

$$G_X(X) = J^{-T} g(q) \quad (3.215)$$

$$\tau = J^T F \quad (3.216)$$

Stability analysis for robot manipulators

- ▶ Rigid-body manipulator dynamics is described by a set of nonlinear equations (see (3.92)), demanding non-linear techniques to analyze stability
- ▶ Lyapunov theory is widely used for nonlinear systems. A scalar function, called Lyapunov function, needs to be found with well defined properties. Typical Lyapunov functions are energy based, but other candidate functions can be tested as well
- ▶ Explicit energy-based techniques can also be used (e.g., Hamiltonian), which are connected to passivity theory. The main idea of the Hamiltonian formulation is to analyze how system energy evolves with time

Lyapunov stability I

- ▶ If all solutions of a dynamic system that start near an equilibrium point x_e stay near x_e over time, the dynamic system is Lyapunov stable around x_e . Additionally, if all solutions that start near x_e converge to x_e , then x_e is asymptotically stable
- ▶ The Lyapunov's first method (the indirect method) linearizes the system around an equilibrium point and checks the eigenvalues of the state-transition matrix to infer stability
- ▶ The Lyapunov's second method (the direct method) is based on a Lyapunov function and does not require linearization
- ▶ Lyapunov's first method determines stability in the immediate vicinity of the equilibrium point and the Lyapunov's second method is more general, enabling to determine how far from the equilibrium point the trajectory can be and still converge over time

Lyapunov stability II

Lyapunov's second method

1. Let x_e be an equilibrium of the nonlinear system $\dot{x} = f(x)$
 2. Let $V(x)$ (Lyapunov function) be a continuously differentiable function such that $V(x_e) = 0$ and $V(x) > 0$ for $x \neq x_e$
 3. if $\|x\| \rightarrow +\infty$ then $V(x) \rightarrow +\infty$
 4. if $\dot{V}(x) < 0$ for $x \neq x_e$, then x_e is globally asymptotically stable
- ▶ Stability conditions of Lyapunov's second method are sufficient but not necessary
 - ▶ There is no systematic method to find Lyapunov functions. A Lyapunov function for a given system is not unique
 - ▶ Lyapunov's second method determines stability without solving the nonlinear differential equation of the system

Lyapunov stability III

- ▶ Lyapunov's second method requires that $\dot{V}(x) < 0$. However, in some cases it is possible to conclude asymptotic stability even when $\dot{V}(x) \leq 0$, thanks to LaSalle's theorem
- ▶ A set M is an (positively) invariant set w.r.t. $\dot{x} = f(x)$, if

$$x(t_0) \in M \Rightarrow x(t) \in M \quad \forall t \geq t_0 \quad (3.217)$$

LaSalle's principle

Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a locally positive definite function such that on the compact set $\Omega_c = \{x \in \mathbb{R}^n : V(x) < c\}$ we have $\dot{V}(x) \leq 0$. Let's define S as

$$S = \{x \in \Omega_c : \dot{V}(x) = 0\}. \quad (3.218)$$

If S contains no invariant sets other than $x = x_e$, then x_e is asymptotically stable

Hamiltonian formulation of robot dynamics

- ▶ The Hamiltonian (H) represents the total energy of the system, corresponding to the sum of potential ($U(q)$) and kinetic ($E(q, \dot{q})$) energies
- ▶ The kinetic energy can be written as

$$E(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \quad (3.219)$$

Alternatively, defining the *generalized momenta* p as a new variable,

$$p = M(q) \dot{q} \quad (3.220)$$

(3.219) can be written as

$$E(p, q) = \frac{1}{2} p^T M(q)^{-1} p \quad (3.221)$$

Therefore,

$$H(p, q) = E(p, q) + U(q) \quad (3.222)$$

Hamiltonian equations I

- ▶ The Hamiltonian equations enable to compute the time derivative of H in straightforward way
- ▶ From (3.222), (3.221), (3.99) and (3.220),

$$\frac{\partial H(p, q)}{\partial p} = \frac{\partial E(p, q)}{\partial p} = M(q)^{-1}p = \dot{q} \quad (3.223)$$

- ▶ Taking now the partial derivative for q ,

$$\frac{\partial H(p, q)}{\partial q} = \frac{\partial E(p, q)}{\partial q} + \frac{\partial U(q)}{\partial q} \quad (3.224)$$

From (3.102) and (3.221),

$$\frac{\partial E(p, q)}{\partial q} = -\frac{1}{2}p^T M(q)^{-1} \frac{d}{dq} (M(q)) M(q)^{-1} p \quad (3.225)$$

Hamiltonian equations II

- Equation (3.225) can be simplified (using (3.220)) to

$$\frac{\partial E(p, q)}{\partial q} = -\frac{1}{2} \dot{q}^T \frac{d}{dq} (M(q)) \dot{q} \quad (3.226)$$

From (3.226) and (3.219),

$$\frac{\partial E(p, q)}{\partial q} = -\frac{\partial E(q, \dot{q})}{\partial q} \quad (3.227)$$

Therefore, (3.224) can be written as

$$\frac{\partial H(p, q)}{\partial q} = -\frac{\partial}{\partial q} (E(q, \dot{q}) - U(q)) \quad (3.228)$$

Hamiltonian equations III

- ▶ Knowing (3.90), (3.228) can be written as

$$\frac{\partial H(p, q)}{\partial q} = -\frac{\partial L(q, \dot{q})}{\partial q} = \tau - \frac{d}{dt} \frac{\partial L(q, \dot{q})}{\partial \dot{q}} \quad (3.229)$$

- ▶ From (3.226) and (3.100), the term

$$\frac{\partial L(q, \dot{q})}{\partial \dot{q}} = \frac{\partial E(q, \dot{q})}{\partial \dot{q}} = M\dot{q} = p \quad (3.230)$$

Therefore, from (3.223), (3.229) and (3.230), the Hamiltonian equations can be written in compact form as

$$\frac{\partial H(p, q)}{\partial p} = \dot{q} \quad (3.231)$$

$$\frac{\partial H(p, q)}{\partial q} = \tau - \dot{p} \quad (3.232)$$

Passive concept I

- $\dot{H}$ is directly obtained from (3.231) and (3.232),

$$\dot{H} = \frac{\partial H(p, q)}{\partial p}^T \dot{p} + \frac{\partial H(p, q)}{\partial q}^T \dot{q} = \dot{q}^T \tau \quad (3.233)$$

- Integrating (3.233) over time gives ($t_1 > t_0$)

$$\int_{t_0}^{t_1} \dot{q}(t)^T \tau(t) dt = H(t_1) - H(t_0) \quad (3.234)$$

- A rigid robot is defined as passive from the input τ to the output $\dot{q}$ if the total energy has a bounded minimum, i.e.,

$$\int_{t_0}^{t_1} \dot{q}(t)^T \tau(t) dt \geq -\beta_H \quad (3.235)$$

with $0 < \beta_H < \infty$. Since $H(t) \geq 0$, (3.234) satisfies (3.235) for $\beta_H = H(t_0)$. Therefore, a rigid robot is passive

Passive concept II

- The result of $\dot{H}$ in (3.233) can also be directly obtained by derivation of (3.222). Knowing (3.219) and (3.101),

$$\dot{H} = \dot{q}^T M(q) \ddot{q} + \frac{1}{2} \dot{q}^T \dot{M}(q) \dot{q} + \dot{q}^T \frac{\partial U}{\partial q} \quad (3.236)$$

From (3.92), (3.93) and (3.94),

$$\dot{H} = \dot{q}^T (\tau - C(q, \dot{q}) \dot{q} - g(q)) + \frac{1}{2} \dot{q}^T \dot{M}(q) \dot{q} + \dot{q}^T g(q) \quad (3.237)$$

which gives

$$\dot{H} = \dot{q}^T \tau + \frac{1}{2} \dot{q}^T (\dot{M}(q) - 2C(q, \dot{q})) \dot{q} \quad (3.238)$$

Using the skew symmetric property of (3.103),

$$\dot{H} = \dot{q}^T \tau \quad (3.239)$$

Interpretation of the passive concept I

- ▶ Equation (3.234) can be written as

$$H(t_0) = H(t_1) + E_d(t_0, t_1) \quad (3.240)$$

with

$$E_d(t_0, t_1) = - \int_{t_0}^{t_1} \dot{q}(t)^T \tau(t) dt \quad (3.241)$$

Since $H(t_0)$ represents the total system energy at the initial time t_0 , and $H(t_1)$ is the total energy at t_1 , the term $E_d(t_0, t_1)$ represents energy dissipation during the time interval $[t_0, t_1]$

- ▶ If the system does not generate energy, $E_d(t_0, t_1)$ cannot be bigger than $H(t_0)$. Therefore,

$$E_d(t_0, t_1) \leq H(t_0) \quad (3.242)$$

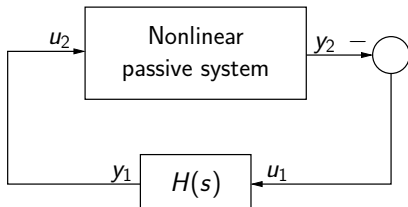
The passivity condition (3.235) follows directly from (3.242) and (3.241)

Interpretation of the passive concept II

- ▶ It can be concluded from (3.242) and (3.235) that a passive system does not generate energy (i.e., it dissipates energy, considering that $E_d(t_0, t_1)$ is not zero)
- ▶ Hence, a very important property of passive systems is that they are always stable. If for a given time instant the external system excitation is turned off, the system dissipates energy as time goes by, avoiding unstable behaviors
- ▶ The passivity condition (3.235) is an elegant way of guaranteeing stability for non-linear (and also linear) control systems
- ▶ Typically, the control law shapes a desired closed-loop Hamiltonian to guarantee passivity from an input control variable (embedded in τ) to the output $\dot{q}$
- ▶ From the mechanical point of view, the interconnection of masses, springs and dampers entails always a passive system, since there are no built-in sources of energy

Linear control of passive systems

Linear control architecture



- ▶ If the nonlinear system is passive, it satisfies the Popov inequality

$$\int_{t_0}^{t_1} y_2(t)^T u_2(t) dt \geq -\gamma_0^2 \quad \text{with} \quad \gamma_0^2 < \infty \quad (3.243)$$

- ▶ The linear control architecture is stable if $H(s)$ is a positive real matrix [LH98]

Stability properties of passive systems

- ▶ A feedback combination of two strictly passive systems is asymptotically stable
- ▶ Any system obtained by a parallel combination of two passive blocks is itself a passive system
- ▶ Any system obtained by a feedback combination of two passive blocks is itself a passive system

Chapter 4

Motion control

This chapter discusses motion control techniques based on computed torque methods [KD02]. Stability analysis [KSL05], [Sam87], singularities and trajectory generation [Yos90] are discussed.

Dynamic modeling

- ▶ Given a set of generalized coordinates q ($n \times 1$) describing robot posture (usually, joint angles for revolute joints), the well-known robot dynamics is given by (see (3.92))

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) = \tau \quad (4.1)$$

- ▶ $M(q)$ is the mass matrix ($n \times n$)
- ▶ n is the number of dof
- ▶ $v(\dot{q}, q)$ represents Coriolis and centripetal forces ($n \times 1$)
- ▶ $g(q)$ is the gravity term ($n \times 1$)
- ▶ τ is the generalized torque acting on q , including friction (τ_f), computer commanded (τ_c) and external torques (τ_e),

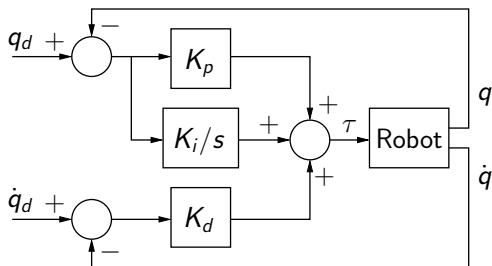
$$\tau = \tau_c + \tau_f + \tau_e \quad (4.2)$$

- ▶ It is assumed throughout the chapter that τ_f and τ_e are negligible, i.e.,

$$\tau \simeq \tau_c \quad (4.3)$$

PID control in the joint space

Control architecture



- ▶ Local decentralized PID control with constant gains implemented for each joint [KD02]
- ▶ Simplicity and low computation cost
- ▶ Dynamic performance varies with robot posture

PID control law I

- ▶ The control law is given by

$$\tau = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + K_i \int_{t_0}^t (q_d - q) d\lambda \quad (4.4)$$

- ▶ $q_d(t)$ and $\dot{q}_d(t)$ represent desired joint positions and velocities
- ▶ K_p , K_d and K_i are $(n \times n)$ positive definite diagonal matrices with diagonal gains K_{pj} , K_{dj} and K_{ij} , respectively
- ▶ The computation of K_{pj} , K_{dj} and K_{ij} for joint j considers the following decentralized model (disturbance free)

$$\tau_j = m_j \ddot{q}_j + F_{vj} \dot{q}_j \quad (4.5)$$

- ▶ $m_j = \max(M_{jj})$ is the maximum magnitude of the (j,j) element of $M(q)$ and F_{vj} represents viscous friction

PID control law II

- ▶ Therefore, from (4.4) and (4.5), the closed loop transfer function for joint j , $H_j(s)$, is given by

$$H_j(s) = \frac{K_{dj}s^2 + K_{pj}s + K_{ij}}{m_js^3 + (K_{dj} + F_{vj})s^2 + K_{pj}s + K_{ij}} \quad (4.6)$$

- ▶ The characteristic polynomial
 $\Delta(s) = m_js^3 + (K_{dj} + F_{vj})s^2 + K_{pj}s + K_{ij}$
- ▶ The common solution in robotics consists of designing a triple pole on the negative real axis, allowing the fastest response without overshoot, i.e.,

$$\Delta(s) = m_j(s + w_j)^3 \quad (4.7)$$

- ▶ The solution is then

$$K_{pj} = 3m_jw_j^2 \quad K_{dj} + F_{vj} = 3m_jw_j \quad K_{ij} = m_jw_j^3 \quad (4.8)$$

Practical issues

- ▶ High gains of K_p and K_d decrease tracking errors, but robustness decreases. Thus, w_j should be kept small compared to the structural resonance frequency w_{rj} (e.g., $w_j = w_{rj}/2$)
- ▶ The predictive action $K_d \dot{q}_d$ reduces tracking errors, although it is often not used
- ▶ In the absence of integral actions, static errors occur due to disturbances including friction, parameter errors and unmodeled dynamics
- ▶ The robot performance based on this PID control scheme is acceptable if high-gear transmission ratios are used, if the robot is moving at low velocities, and if high position gains are assigned

Lyapunov stability for PD control in the joint space I

- ▶ Let's analyze now Lyapunov stability for a PD control plus gravity compensation. The torque τ is given by

$$\tau = g(q) + K_p(q_d - q) - K_d\dot{q} \quad (4.9)$$

where q_d is a constant desired joint position, and K_p and K_d are positive definite matrices

- ▶ From (4.1) and (4.9), the overall nonlinear system can be written in the form $\dot{x} = f(x)$ as

$$\frac{\partial}{\partial t} \begin{bmatrix} \tilde{q} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} -\dot{q} \\ M(q)^{-1} (K_p \tilde{q} - K_d \dot{q} - v(q, \dot{q})) \end{bmatrix} \quad (4.10)$$

where

$$\tilde{q} = q_d - q \quad (4.11)$$

- ▶ Note that the variable q in (4.10) can be substituted by $\tilde{q}$ using (4.11)

Lyapunov stability for PD control in the joint space II

- ▶ The candidate Lyapunov function $V(\tilde{q}, \dot{q})$ is given by

$$V(\tilde{q}, \dot{q}) = \frac{1}{2} \dot{q}^T M(q_d - \tilde{q}) \dot{q} + \frac{1}{2} \tilde{q}^T K_p \tilde{q} \quad (4.12)$$

- ▶ $V(\tilde{q}, \dot{q}) > 0$ (except at the origin where it is zero), since M and K_p are positive definite matrices
- ▶ The time derivative of $V(\tilde{q}, \dot{q})$ is given by

$$\dot{V} = \dot{q}^T M(q) \ddot{q} + \frac{1}{2} \dot{q}^T \dot{M}(q) \dot{q} + \tilde{q}^T K_p \dot{\tilde{q}} \quad (4.13)$$

which can be written as

$$\dot{V} = \frac{1}{2} \dot{q}^T \left[\dot{M}(q) - 2C(q, \dot{q}) \right] \dot{q} + \dot{q}^T K_p \tilde{q} - \dot{q}^T K_d \dot{q} + \tilde{q}^T K_p \dot{\tilde{q}} \quad (4.14)$$

Lyapunov stability for PD control in the joint space III

- ▶ Manipulating (4.14) and using the skew symmetric property (3.103), we get

$$\dot{V}(\tilde{q}, \dot{q}) = -\dot{q}^T K_d \dot{q} \leq 0 \quad (4.15)$$

- ▶ Using LaSalle's principle, the set S (see (3.218)) corresponds to

$$S = \left\{ x = \begin{bmatrix} \tilde{q} \\ \dot{q} \end{bmatrix} \in \Omega_c : \dot{V}(\tilde{q}, \dot{q}) = 0 \right\} = \{ \forall \tilde{q}, \dot{q} = 0 \} \quad (4.16)$$

- ▶ The only initial condition in S for which $x(t) \in S$ for all t is when $\tilde{q} = 0$ and $\dot{q} = 0$. Therefore, $\dot{V}(\tilde{q}, \dot{q}) < 0$ except for $\tilde{q} = 0$, where $\dot{V}(\tilde{q}, \dot{q}) = 0$. Hence, global asymptotic stability of the origin exists, i.e.,

$$\lim_{t \rightarrow \infty} q(t) = q_d \quad (4.17)$$

Lyapunov stability for PD control in the joint space IV

- ▶ Let's get more insight about the only initial condition that satisfies $\dot{V}(\tilde{q}, \dot{q}) = 0$
- ▶ If $\dot{q} = 0$, then from (4.1) and (4.9),

$$M(q)\ddot{q} = K_p\tilde{q} \quad (4.18)$$

- ▶ If

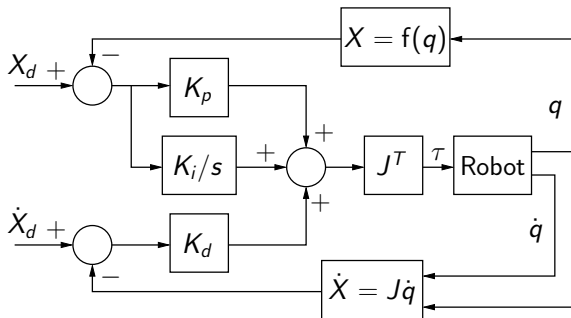
$$\tilde{q} \neq 0 \Rightarrow \ddot{q} \neq 0 \Rightarrow \dot{q} \neq 0 \quad (4.19)$$

which violates the condition $\dot{q} = 0$

- ▶ Therefore, the only equilibrium point where $\dot{q} = 0$ is for $\tilde{q} = 0$, i.e., $q = q_d$
- ▶ Stability of PID based controllers using Lyapunov theory is more elaborate, and can be seen in [KSL05]

PID control in the task space I

Task space control



- ▶ When the motion is specified in the task space (represented by the vector X), one of the following schemes can be used
 - ▶ The control law is designed in the task space
 - ▶ The task space trajectory is mapped to the joint space for control

PID control in the task space II

Joint space control

- ▶ Desired task trajectory can be mapped to the joint space
 - ▶ Using inverse geometric models to compute joint positions from task positions

$$q_d = f^{-1}(X_d) \quad (4.20)$$

- ▶ Using inverse kinematics models to compute joint velocities from task velocities,

$$\dot{q}_d = J(q_d)^{-1} \dot{X}_d \quad (4.21)$$

- ▶ For redundant robots, the matrix $J(q_d)^{-1}$ should be replaced by the Moore-Penrose pseudoinverse $J^\#(q_d)$,

$$J^\#(q_d) = J^T(q_d) \left(J(q_d) J^T(q_d) \right)^{-1} \quad (4.22)$$

- ▶ Other Jacobian pseudoinverses can also be used

Feedback linearization and decoupled control

- ▶ When the task requires high dynamic accuracy regardless of robot postures, it is necessary to include dynamic models in control design
- ▶ Feedback linearization and decoupled control is based on canceling nonlinearities of robot dynamics to guarantee consistency in dynamic responses
- ▶ Such control is known as computed torque control or inverse dynamics control, where linearization and decoupling is based on computed (and applied) torques, taking into account dynamic models
- ▶ The idea is to transform a nonlinear control problem into a linear one by using an appropriate feedback law

Computed torque control in the joint space

Feedback linearization

- ▶ Setting the computed torque τ_c equal to (see (4.2))

$$\tau_c = \hat{M}(q)w + \hat{v}(\dot{q}, q) + \hat{g}(q) \quad (4.23)$$

where $(\hat{\cdot})$ means estimate, and neglecting estimation errors, friction and external torques, (4.1) becomes

$$\ddot{q} = w \quad (4.24)$$

- ▶ w is the new control variable
- ▶ w can be defined for
 - ▶ Tracking control when the trajectory is defined
 - ▶ Position control when the final point is defined

Trajectory control in the joint space I

- ▶ Defining w as

$$w = \ddot{q}_d + K_d(\dot{q}_d - \dot{q}) + K_p(q_d - q) \quad (4.25)$$

where K_d and K_p are positive definite diagonal matrices, the following decoupled error dynamics is obtained (see (4.24))

$$\ddot{e} + K_d\dot{e} + K_p e = 0 \quad (4.26)$$

with

$$e = q_d - q \quad (4.27)$$

- ▶ A critically damped error dynamics with natural frequency w_j entails for each joint j the gains

$$K_{pj} = w_j^2 \quad (4.28)$$

$$K_{dj} = 2w_j \quad (4.29)$$

Trajectory control in the joint space II

- ▶ Taking into account estimation errors in $M(q)$, $v(q, \dot{q})$ and $g(q)$, and knowing (4.1), (4.23), and (4.25), the error dynamics becomes

$$\ddot{e} + K_d \dot{e} + K_p e = \hat{M}(q)^{-1} \Delta E \quad (4.30)$$

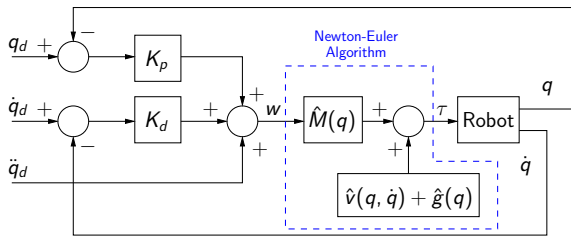
with

$$\Delta E = \left(M(q) - \hat{M}(q) \right) \ddot{q} + v(q, \dot{q}) - \hat{v}(q, \dot{q}) + g(q) - \hat{g}(q) \quad (4.31)$$

- ▶ Estimation errors constitute an excitation for the error equation. When modeling errors are big, it is necessary to increase control gains, without jeopardizing stability margins [Sam87]

Trajectory control in the joint space III

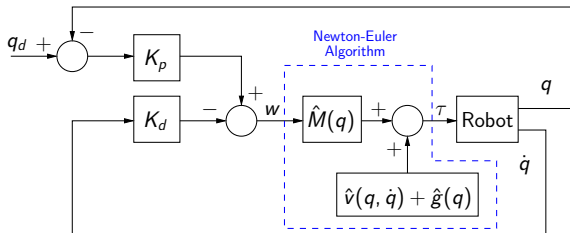
Tracking control scheme



- ▶ PD computed torque control in the joint space, based on realtime dynamic model computation
- ▶ The computation of $\hat{M}(q)$, $\hat{v}(q, \dot{q})$ and $\hat{g}(q)$ can be done using the recursive Newton-Euler algorithm, Euler-Lagrange algorithm or a combination of both

Position control in the joint space

Position control scheme



► Making

$$w = -K_d \dot{q} + K_p(q_d - q) \quad (4.32)$$

the closed-loop system dynamics becomes

$$\ddot{q} + K_d \dot{q} + K_p q = K_p q_d \quad (4.33)$$

- This solution is globally exponentially stable by properly choosing K_p and K_d

Computed torque control in the task space

- ▶ Dynamic control in the task space is also known as resolved acceleration control
- ▶ Knowing that the Jacobian J relates task and joint velocities as

$$\dot{X} = J\dot{q} \quad (4.34)$$

Equation (4.1) can be written as

$$M(q)J^{-1}(\ddot{X} - \dot{J}\dot{q}) + v(\dot{q}, q) + g(q) = \tau \quad (4.35)$$

Therefore, a control law that linearizes and decouples the task space equations is given by

$$\hat{M}(q)J^{-1}(w - \dot{J}\dot{q}) + \hat{v}(\dot{q}, q) + \hat{g}(q) = \tau_c \quad (4.36)$$

entailing without estimation errors (and neglecting or compensating friction and external torques)

$$w = \ddot{X} \quad (4.37)$$

Trajectory control in the task space I

- From (4.37), several control schemes can be defined. For PD control, w is given by

$$w = \ddot{X}_d + K_d(\dot{X}_d - \dot{X}) + K_p(X_d - X) \quad (4.38)$$

entailing the decoupled error dynamics (similar analysis for the joint space case)

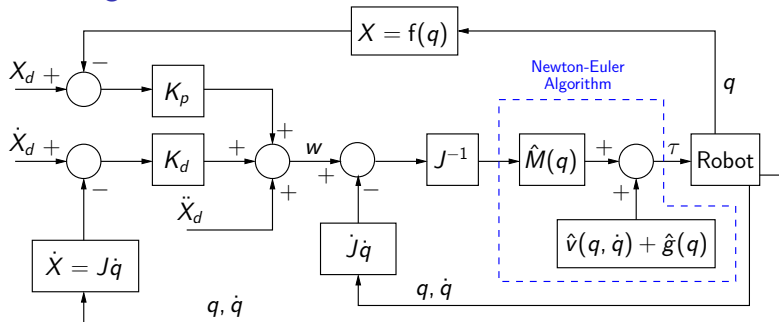
$$\ddot{e}_x + K_d \dot{e}_x + K_p e_x = 0 \quad (4.39)$$

with

$$e_x = X_d - X \quad (4.40)$$

Trajectory control in the task space II

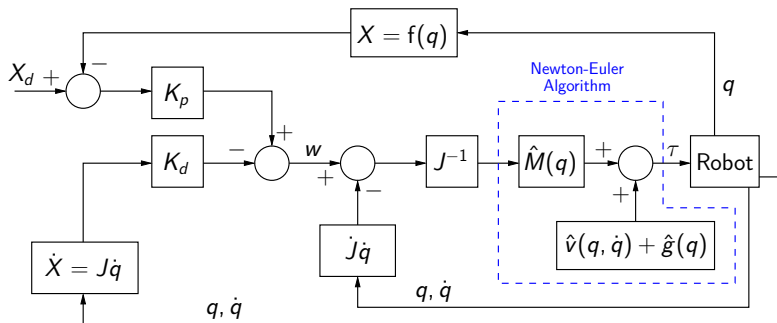
Tracking control scheme



- ▶ This control algorithm assumes that the robot is non-redundant due to the computation of J^{-1}
- ▶ For redundant robots, J^{-1} can be substituted by the dynamically consistent generalized inverse $J^\#$ (see (3.210)) or other task space schemes can be applied (see impedance control for redundant robots, pp. 211-213)

Position control in the task space

Position control scheme



- The position control for the task space has a control architecture similar to trajectory control, where the references for velocity and acceleration are removed (the same applies for the joint space case)

Passivity-based position control I

- ▶ To converge to a desired q_d , the energy minimum of the open-loop system at $q = 0$, $\dot{q} = 0$ has to shift to the energy minimum of the closed loop system at $q = q_d$, $\dot{q} = 0$
- ▶ This shifting can be obtained by reshaping the potential energy of the system, only. The kinetic energy does not have to be reshaped since its minimum is for $\dot{q} = 0$
- ▶ Let $U^*(q)$ be the desired potential energy of the closed loop system. By defining the control law as

$$\tau = -\frac{\partial U^*(q)}{\partial q} + \frac{\partial U(q)}{\partial q} + v \quad (4.41)$$

where v is the new control variable, the Hamiltonian equations can be obtained

Passivity-based position control II

- From (3.231) and (3.232),

$$\dot{q} = \frac{\partial E(p, q)}{\partial p} \quad (4.42)$$

$$\dot{p} = -\frac{\partial E(p, q)}{\partial q} - \frac{\partial U^*(q)}{\partial q} + v \quad (4.43)$$

- Comparing (4.42)-(4.43) with (3.231)-(3.232) it can be inferred that the Hamiltonian of the closed loop system $H^*(p, q)$ is represented by

$$H^*(p, q) = E(p, q) + U^*(q) \quad (4.44)$$

which satisfies (see (3.233))

$$\dot{H}^* = \dot{q}^T v \quad (4.45)$$

Passivity-based position control III

- ▶ This implies that the system is passive from the new input v to the output $\dot{q}$
- ▶ Making

$$v = -K_d \dot{q} \quad (4.46)$$

where $K_d > 0$ is a diagonal matrix, (4.45) becomes

$$\dot{H}^* = -\dot{q}^T K_d \dot{q} \quad (4.47)$$

Various choices are possible for the desired potential energy $U^*(q)$. An obvious one with a strict minimum at $q = q_d$ is

$$U^*(q) = \frac{1}{2}(q_d - q)^T K_p (q_d - q) \quad (4.48)$$

where $K_p > 0$ is a diagonal matrix

Passivity-based position control IV

- From (4.41), (4.46), (4.48) and (3.94),

$$\tau = K_p(q_d - q) - K_d\dot{q} + g(q) \quad (4.49)$$

- This solution represents a PD control in joint space with gravity compensation
- Let's analyze global asymptotic stability. Equation (4.47) is negative semi-definite. This means that the equilibrium point $\dot{q} = 0$, with $q \neq q_d$ is a possible solution. To show that this cannot happen, LaSalle's theorem can be used. Suppose that $\dot{H}^* \equiv 0$. Then, $\dot{q} \equiv 0$, and hence $\ddot{q} \equiv 0$. The overall system (see (4.1) and (4.49)) is represented by

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) = K_p(q_d - q) - K_d\dot{q} + g(q) \quad (4.50)$$

- With $\dot{q} = 0$ and $\ddot{q} = 0$, (4.50) becomes

$$q = q_d \quad (4.51)$$

Therefore, global asymptotic stability has been shown

Stability of null space motion I

- For redundant robots, the overall power P_A can be computed by the general $\dot{q}$ solution (see (3.156)) and by the general τ solution (see (3.192)). It is given by

$$P_A = \dot{q}^T \tau = \dot{q}^T J^T J^\# T \tau + \dot{q}_0^T \left(I - J^T J^\# T \right) \tau_0 = P_t + P_N \quad (4.52)$$

where P_N is the null space power

- P_N can be written in different ways, such as

$$P_N = \dot{q}^T \left(I - J^T J^\# T \right) \tau_0 = \dot{q}_0^T \left(I - J^T J^\# T \right) \tau \quad (4.53)$$

The proof of (4.53) can be obtained by direct substitution of τ and $\dot{q}$ by (3.192) and (3.156), respectively

Stability of null space motion II

- ▶ To guarantee stability of the null space motion, τ_0 can be defined as

$$\tau_0 = -\alpha M(q)\dot{q} \quad (4.54)$$

where α is a positive constant or a diagonal matrix with positive constants

- ▶ From (3.169), the matrix

$$\left(I - J^T J^{\#T}\right) M \quad (4.55)$$

is non-negative definite

- ▶ Hence, the term

$$\dot{q}^T \left[-\alpha \left(I - J^T J^{\#T}\right) M \right] \dot{q} \leq 0 \quad (4.56)$$

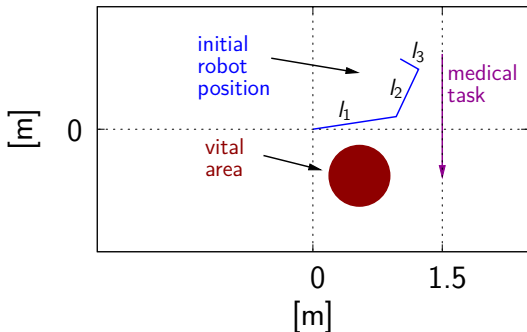
entailing stability of the null space motion

Trajectory generation I

- ▶ Trajectory generation is left to the physician for task execution. In case of autonomous motion (e.g., image guided motion), sensory information is mapped into desired task motion
- ▶ Robot redundancy can be exploited in multiple ways through null-space motion design
- ▶ Null-space velocities or null-space torques can be designed through $\dot{q}_0$ or τ_0 , respectively (see (3.156) and (3.192)), to tackle some criterion such as maximizing robot manipulability, avoiding vital organ contacts and keeping a desired posture
- ▶ One strategy is to design vectors $\dot{q}_0$ or τ_0 to be collinear with posture errors (i.e., pointing to the corrective direction)

Trajectory generation II

Straight line motion avoiding obstacles



- Let's analyze an example with a three-link planar robot with link lengths l_1 , l_2 and l_3 , where the physician has to perform a straight line motion, avoiding vital areas

Trajectory generation III

- For our robot, $l_1 = 1$ [m], $l_2 = 1$ [m] and $l_3 = 0.3$ [m] and the initial joint position q_i is given by

$$q_i = \begin{bmatrix} \pi/12 \\ \pi/3 \\ \pi/3 \end{bmatrix} \quad (4.57)$$

Therefore, the forward geometric model can be computed by

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} l_1 \cos(q_1) + l_2 \cos(q_1 + q_2) + l_3 \cos(q_1 + q_2 + q_3) \\ l_1 \sin(q_1) + l_2 \sin(q_1 + q_2) + l_3 \sin(q_1 + q_2 + q_3) \end{bmatrix} \quad (4.58)$$

where x and y represent the end-effector Cartesian position and q_1 , q_2 and q_3 are joint angles (starting from the base). In compact form, (4.58) can be written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} l_1 C_1 + l_2 C_{12} + l_3 C_{123} \\ l_1 S_1 + l_2 S_{12} + l_3 S_{123} \end{bmatrix} \quad (4.59)$$

Trajectory generation IV

- From (4.59), the end-effector Jacobian J is given by

$$J = \begin{bmatrix} -l_1 S_1 - l_2 S_{12} - l_3 S_{123} & -l_2 S_{12} - l_3 S_{123} & -l_3 S_{123} \\ l_1 C_1 + l_2 C_{12} + l_3 C_{123} & l_2 C_{12} + l_3 C_{123} & l_3 C_{123} \end{bmatrix} \quad (4.60)$$

- The task position error can be converted into a desired task velocity $\dot{X}_d$,

$$\dot{X}_d = K_p (X_d - X) \quad (4.61)$$

where X_d is the task position controlled by the surgeon, K_p is a positive gain and X is the current robot position

Trajectory generation V

- ▶ Defining now the null space velocity $\dot{q}_0$ as a vector pointing to the position error w.r.t. a given desired posture q_{dp} ,

$$\dot{q}_0 = K_q (q_{dp} - q) \quad (4.62)$$

where K_q is a positive constant, the desired joint position q_d can be obtained by integrating $\dot{q}_d$ given by (see (3.155) and (3.156))

$$\dot{q}_d = J^\# \dot{X}_d + (I - J^\# J) \dot{q}_0 \quad (4.63)$$

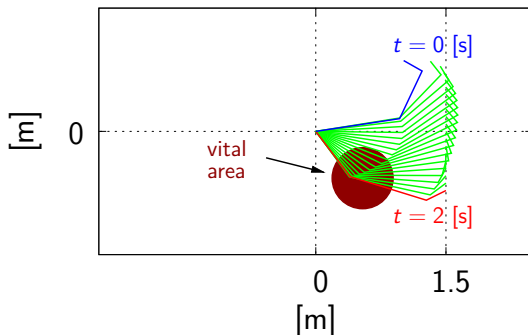
with

$$J^\# = J^T (J J^T)^{-1} \quad (4.64)$$

- ▶ Note that with this approach there is no need to know robot dynamics to compute joint trajectories. Knowing $\dot{q}_d$, there are multiple control architectures for joint control

Trajectory generation VI

Results without null space design

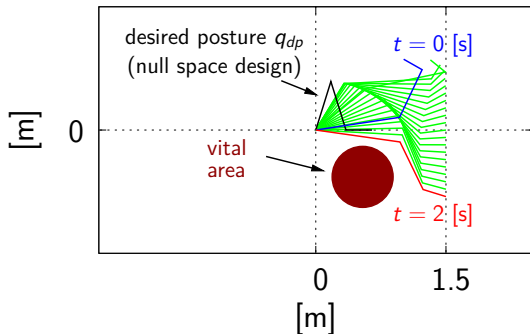


- This picture shows results for $K_p = 10$, $K_q = 0$, corresponding to no null space design, i.e.,

$$\dot{q}_0 = 0 \quad (4.65)$$

Trajectory generation VII

Results with null space design I

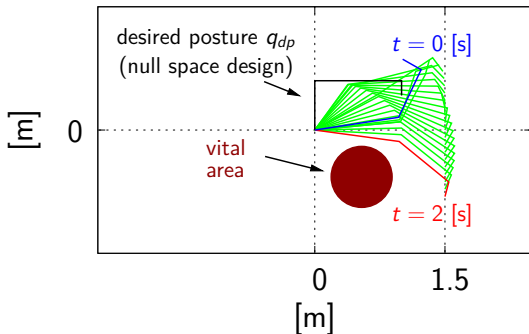


- This picture shows results for $K_p = 10$, $K_q = 10$, and

$$q_{dp} = \begin{bmatrix} 80\pi/180 & -160\pi/180 & 80\pi/180 \end{bmatrix}^T \quad (4.66)$$

Trajectory generation VIII

Results with null space design II



- This picture shows results for $K_p = 10$, $K_q = 10$, and

$$q_{dp} = \begin{bmatrix} \pi/2 & -\pi/2 & -\pi/2 \end{bmatrix}^T \quad (4.67)$$

Trajectory generation IX

- ▶ Singularities is a serious problem in robotics. A singularity configuration of a general n -link manipulator exists when the end-effector cannot move in certain directions. Several techniques can tackle this problem
- ▶ From the robot design perspective, task singularities can be mitigated by increasing robot redundancy
- ▶ From the mathematical perspective, singularities can be detected when the Jacobian J loses rank. In other words, $\det(J) = 0$ for non-redundant robots, or $\det(JJ^T) = 0$ for redundant robots
- ▶ From the computational perspective, numerical problems show up near singularity configurations, entailing divisions by zero (or close to zero) when computing J^{-1} or $(JJ^T)^{-1}$
- ▶ From the trajectory generation perspective, singularity distance measures can be used to keep away from singular configurations

Trajectory generation X

- ▶ Away from singularities, (4.63) can be computed in two steps [Yos90], i.e.,

1. Compute α from

$$(JJ^T)\alpha = \dot{X}_d - J\dot{q}_0 \quad (4.68)$$

2. Compute $\dot{q}_d$ from

$$\dot{q}_d = J^T\alpha + \dot{q}_0 \quad (4.69)$$

- ▶ Close to singularities, the procedure can be generalized to

1. Compute α from

$$(JJ^T + k_a I)\alpha = \dot{X}_d - J\dot{q}_0 \quad (4.70)$$

2. Compute $\dot{q}_d$ from

$$\dot{q}_d = J^T\alpha + \dot{q}_0 \quad (4.71)$$

where k_a is non-negative. For instance, k_a can be given by

$$k_a = \begin{cases} \beta_0(1 - w/w_0)^2, & w < w_0 \\ 0, & w \geq w_0 \end{cases} \quad (4.72)$$

where $\beta_0 > 0$, $w_0 > 0$ and w is a singularity distance measure

Trajectory generation XI

- ▶ Let's analyze the behavior of the three-link robot close to a singularity, i.e., for a medical task making a downward vertical motion at the abscissa 0.3 [m] when the robot initial position q_i is given by

$$q_i = \begin{bmatrix} \pi & -170\pi/180 & -10\pi/180 \end{bmatrix}^T \quad (4.73)$$

- ▶ For the null space design, $\dot{q}_0$ is given by (see (3.161))

$$\dot{q}_0 = K_0 \frac{\partial \gamma(q)}{\partial q} \quad (4.74)$$

with

$$K_0 = 10 \quad (4.75)$$

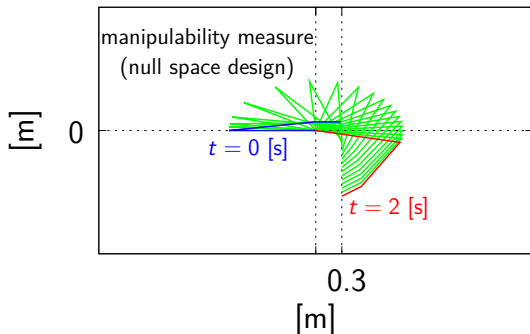
and

$$\gamma(q) = \det(JJ^T) \quad (4.76)$$

- ▶ Equation (4.76) is just a slight variation of (3.162), where the square root has been eliminated

Trajectory generation XII

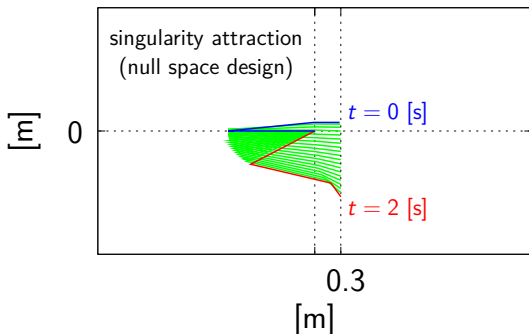
Avoiding singularities with null-space design



- This picture shows results for $\dot{q}_d$ given by (4.63), with $K_p = 10$ (see (4.61)) and $\dot{q}_0$ computed from (4.74)

Trajectory generation XIII

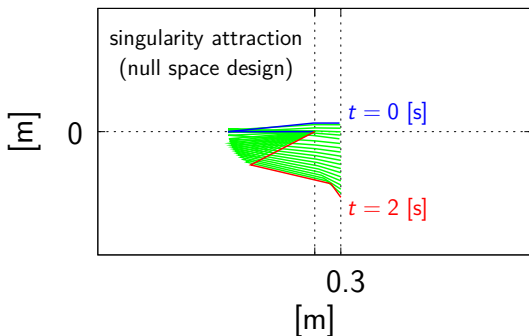
Attracting singularities with null-space design I



- This picture shows results for $\dot{q}_d$ given by (4.63) and $K_0 = -10$ (highly risky due to divisions close to zero)

Trajectory generation XIV

Attracting singularities with null-space design II



- This picture shows results for $\dot{q}_d$ given by (4.70)-(4.72), with $\beta_0 = 0.2$, $w_0 = 0.03$ and $K_0 = -10$

Chapter 5

Interaction control

This chapter discusses interaction control techniques based on computed torque methods. Active stiffness control [Sal80], impedance control [Hog85], and explicit force control [VK93] are analyzed.

Interaction control problem

- ▶ Interaction control is fundamentally different from motion control due to the presence of interaction forces
- ▶ Interaction forces change the overall system dynamics, which becomes a function of poorly known (or unknown) contact parameters
- ▶ Interaction modeling might have wide parameter changes due to free-space and contact states, demanding robust and/or adaptive control techniques
- ▶ Motion control and force control can co-exist for interaction tasks
- ▶ Force signals are usually more noisy than position signals, creating extra problems for control
- ▶ The main paradigms of surgical robotics, telemanipulation and comanipulation, require interaction control architectures [SCQ09]

Applications

Robotic-assisted minimally invasive surgery

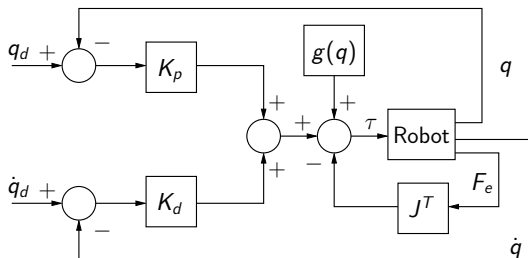
- ▶ Non damaging tissue manipulation requires accuracy, safety and force control
- ▶ Guaranteed stability in contact tasks involving heterogeneous objects
- ▶ Environment parameter estimation for force control (or alternatively, passive-based control)
- ▶ Multiple contact control (e.g., distal and trocar contacts)

Robots for telediagnosis

- ▶ Compliant motion control with and without force sensing
- ▶ Soft-robotics (e.g., safe patient-robot interactions during tele-echography)

Active stiffness control I

Active stiffness control scheme



- Control scheme proposed by Salisbury [Sal80]. The stiffness formulation in the compliant frame is represented by a diagonal stiffness K_c that relates robot forces with differential displacements around the equilibrium point, i.e.,

$$F_e = K_c dX_c = K_c J dq \quad (5.1)$$

Active stiffness control II

- ▶ F_e represents the force that the robot applies on the environment. The term $-J^T F_e$ represents the torque that the environment applies on the robot (Newton's third law)
- ▶ Knowing (3.216) and (5.1), a joint stiffness matrix K_p can be defined as

$$K_p = J^T K_c J \quad (5.2)$$

- ▶ K_p is not diagonal but symmetric. A control law based on (5.2) can be written as

$$\tau_c = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + g(q) \quad (5.3)$$

where K_d is a diagonal damping matrix and $g(q)$ compensates gravity forces

Active stiffness control III

- ▶ Let's analyze the equilibrium point. Neglecting friction terms, the overall dynamic equation in contact is given by (see (3.92))

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) = \tau_c - J^T F_e \quad (5.4)$$

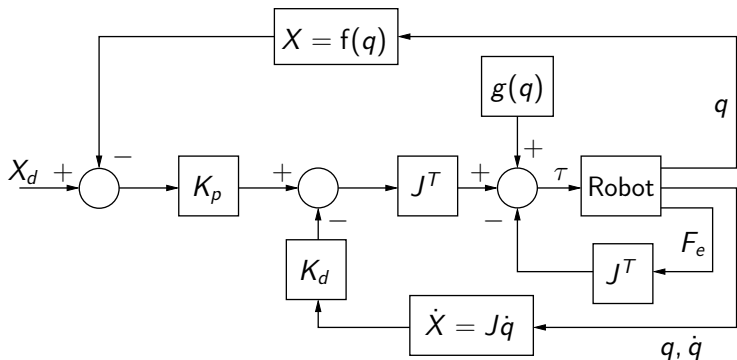
For $\dot{q}$ and $\ddot{q}$ equal to zero, (5.4) and (5.3) gives

$$K_p(q_d - q) = J^T F_e \quad (5.5)$$

- ▶ From (5.2) and (5.5), (5.1) is obtained for small displacements around equilibrium (i.e., $q_d - q \simeq dq$)
- ▶ Compliance control is similar to active stiffness control, and impedance control generalizes the approach. All these techniques manage the interaction force via motion control (reference inputs are position-based signals), without explicit force feedback

Compliance control I

Compliance control scheme



- Compliance control in the task space with gravity compensation [SS00]

Compliance control II

- ▶ The compliance control law is given by

$$\tau_c = J^T \left[K_p(X_d - X) - K_d\dot{X} \right] + g(q) \quad (5.6)$$

- ▶ Therefore, the overall dynamics in contact can be written as

$$M(q)\ddot{q} + v(\dot{q}, q) = J^T \left[K_p(X_d - X) - K_d\dot{X} \right] - J^T F_e \quad (5.7)$$

- ▶ In equilibrium ($\dot{q} = 0$ and $\ddot{q} = 0$), (5.7) becomes

$$K_p(X_d - X_\infty) = F_e \quad (5.8)$$

with $X_\infty \geq X_e$. X_∞ and X_e represent respectively the contact position in equilibrium and the environment position

Compliance control III

- Modeling the environment by a spring K_e ,

$$F_e = K_e(X_\infty - X_e) \quad (5.9)$$

From (5.8) and (5.9),

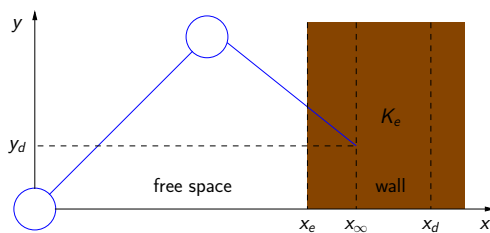
$$X_\infty = (K_e + K_p)^{-1}(K_p X_d + K_e X_e) \quad (5.10)$$

Additionally, for diagonal matrices K_e and K_p , (5.10) and (5.9) gives the steady-state force F_e ,

$$F_e = (K_e + K_p)^{-1} K_e K_p (X_d - X_e) \quad (5.11)$$

Compliance control IV

Example: Two-link planar arm in contact with an elastic plane



► For this example, the environment and control matrices are

$$K_e = \begin{bmatrix} k_{e,x} & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad K_p = \begin{bmatrix} k_{p,x} & 0 \\ 0 & k_{p,y} \end{bmatrix} \quad (5.12)$$

Compliance control V

- The position vectors are

$$X_e = \begin{bmatrix} x_e \\ \forall y \end{bmatrix} \quad X_d = \begin{bmatrix} x_d \\ y_d \end{bmatrix} \quad \text{and} \quad X_\infty = \begin{bmatrix} x_\infty \\ y_\infty \end{bmatrix} \quad (5.13)$$

Therefore, X_∞ is given by (see (5.10)),

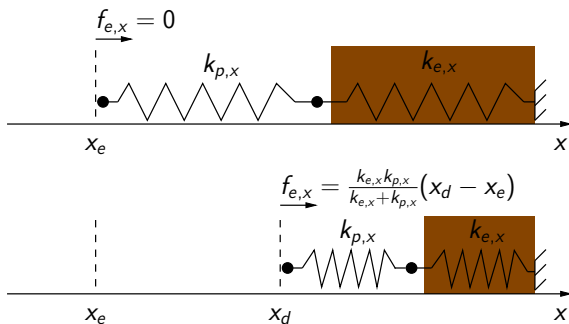
$$X_\infty = \begin{bmatrix} \frac{k_{p,x}x_d + k_{e,x}x_e}{k_{e,x} + k_{p,x}} \\ y_d \end{bmatrix} \quad (5.14)$$

For F_e (see (5.11)),

$$F_e \equiv \begin{bmatrix} f_{e,x} \\ f_{e,y} \end{bmatrix} = \begin{bmatrix} \frac{k_{e,x}k_{p,x}}{k_{e,x} + k_{p,x}}(x_d - x_e) \\ 0 \end{bmatrix} \quad (5.15)$$

Compliance control VI

Springs in series



- The result of (5.15) can be interpreted as the interconnection of two springs in series. The robot behaves like a spring, serially connected to the environment

Compliance control VII

- If $k_{p,x} \gg k_{e,x}$, then (see (5.14) and (5.15))

$$X_\infty \simeq \begin{bmatrix} x_d \\ y_d \end{bmatrix} \quad \text{and} \quad F_e \simeq \begin{bmatrix} k_{e,x}(x_d - x_e) \\ 0 \end{bmatrix} \quad (5.16)$$

The elastic forces are mainly due to the environment (passive compliance)

- If $k_{p,x} \ll k_{e,x}$, then (see (5.14) and (5.15))

$$X_\infty \simeq \begin{bmatrix} x_e \\ y_d \end{bmatrix} \quad \text{and} \quad F_e \simeq \begin{bmatrix} k_{p,x}(x_d - x_e) \\ 0 \end{bmatrix} \quad (5.17)$$

The elastic forces are mainly generated by robot control gains (active compliance)

Impedance control I

- ▶ According to Hogan [Hog85], the idea of impedance control is to assign a prescribed robot dynamic behavior in the presence of external interactions, matching the dynamics of mass-spring-damper systems
- ▶ The end-effector velocity $\dot{X}$ and the robot applied force F_e are related by a mechanical impedance Z . In the Laplace domain,

$$-F_e(s) = Z(s)\dot{X}(s) \quad (5.18)$$

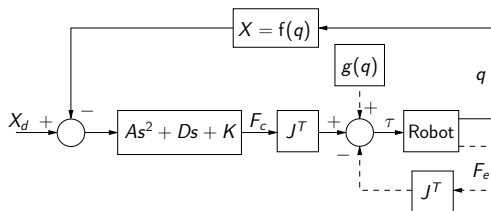
with

$$sZ(s) = As^2 + Ds + K \quad (5.19)$$

where A , K and D are the parameters of a mass-spring-damper system, respectively

Impedance control II

Impedance control scheme in the task space without force sensing



- The overall dynamics in contact can be written as

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) = \tau_c - J^T F_e \quad (5.20)$$

with

$$\tau_c = J^T \left[A(\ddot{X}_d - \ddot{X}) + D(\dot{X}_d - \dot{X}) + K(X_d - X) \right] + g(q) \quad (5.21)$$

Impedance control III

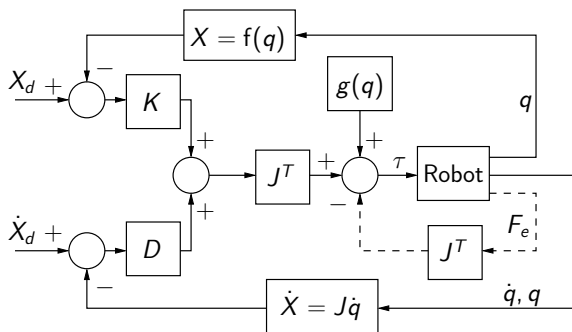
- ▶ Neglecting robot dynamic effects, (5.20) and (5.21) gives

$$A(\ddot{X}_d - \ddot{X}) + D(\dot{X}_d - \dot{X}) + K(X_d - X) \approx F_e \quad (5.22)$$

- ▶ In steady-state, (5.22) is equivalent to compliance control (see (5.8)), where K plays the role of K_p
- ▶ τ_c in (5.21) can also compensate for dynamic terms (e.g., gravity and Coriolis)
- ▶ If there is no contact, the robot under impedance control generates a force F_c according to the mass-spring-damper system, based on X , $\dot{X}$ and $\ddot{X}$ displacements around the equilibrium point

Impedance control IV

Impedance control scheme in the task space without force sensing and without inertia shaping



- ▶ The mass term in (5.19) is hard to implement in practice. Often, impedance control takes into account spring-damper terms only

Impedance control V

- ▶ Using force sensing and the full robot dynamics model, an impedance control scheme can be derived
- ▶ Starting from the robot dynamics in contact,

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) + J^T F_e = \tau_c \quad (5.23)$$

the desired impedance behavior is given by (see (5.18) and (5.19))

$$A(\ddot{X}_d - \ddot{X}) + D(\dot{X}_d - \dot{X}) + K(X_d - X) = F_e \quad (5.24)$$

Equation (5.24) can be written as

$$\ddot{X} = \ddot{X}_d + A^{-1} \left[D(\dot{X}_d - \dot{X}) + K(X_d - X) - F_e \right] \quad (5.25)$$

Impedance control VI

- Therefore, using a procedure similar to (4.35)-(4.38), corresponding to resolved acceleration control, τ_c in (5.23) can shift to another control variable w by

$$\tau_c = MJ^{-1}(w - \dot{J}\dot{q}) + v(\dot{q}, q) + g(q) + J^T F_e \quad (5.26)$$

with (see (5.25))

$$w = \ddot{X}_d + A^{-1} \left[D(\dot{X}_d - \dot{X}) + K(X_d - X) - F_e \right] \quad (5.27)$$

guaranteeing the desired behavior of (5.24)

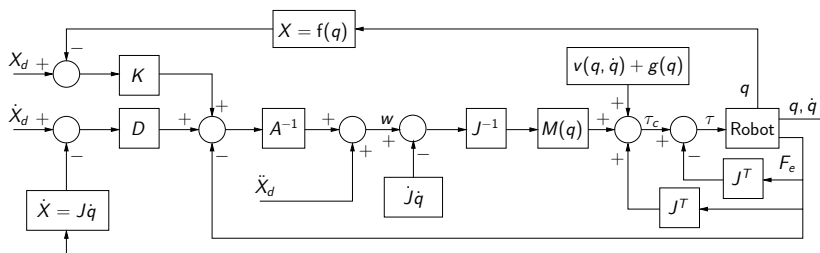
- If

$$A^{-1} = JM^{-1}J^T \quad (5.28)$$

there is no inertia shaping in the task space. In this case, the F_e terms in (5.26) and (5.27) are eliminated (no force feedback is needed)

Impedance control VII

Impedance control scheme in the task space with force sensing



- ▶ General impedance control scheme with force feedback based on (5.26) and (5.27)
- ▶ This control scheme requires the computation of J^{-1} , which is not possible for redundant robots

Impedance control VIII

- ▶ From the control point of view, a redundant robot means that it has more DOF than the task space
- ▶ For redundant robots, the following control scheme in the task space can be implemented
- ▶ Starting from the robot dynamics in contact,

$$M(q)\ddot{q} + v(\dot{q}, q) + g(q) + J^T F_e = \tau_c \quad (5.29)$$

and pre-compensating $v(\dot{q}, q)$, $g(q)$ and $J^T F_e$,

$$M(q)\ddot{q} = \tau' \quad (5.30)$$

with

$$\tau_c = \tau' + v(\dot{q}, q) + g(q) + J^T F_e \quad (5.31)$$

- ▶ Shifting $M(q)$ to the right side of (5.30), and multiplying by J ,

$$J\ddot{q} = JM(q)^{-1}\tau' \quad (5.32)$$

Impedance control IX

- ▶ Since

$$\dot{X} = J\dot{q} \rightarrow J\ddot{q} = \ddot{X} - \dot{J}\dot{q} \quad (5.33)$$

and applying an additional Cartesian force F_c through τ' ,
(5.32) becomes

$$\ddot{X} - \dot{J}\dot{q} = JM(q)^{-1}J^TF_c \quad (5.34)$$

- ▶ The inertia matrix of the task space $\Lambda(q)$ has the following inverse

$$\Lambda(q)^{-1} = JM(q)^{-1}J^T \quad (5.35)$$

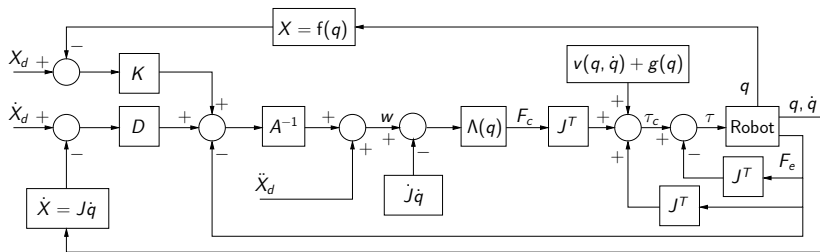
which always exists, even if J is non-square ($M(q)^{-1}$ always exists)

- ▶ Therefore, from (5.34) and (5.35), the dynamic equation in the task space can be written as

$$\Lambda(q)\ddot{X} - \Lambda(q)\dot{J}\dot{q} = F_c \quad (5.36)$$

Impedance control X

Impedance control scheme in the task space with force sensing for redundant robots



► Making

$$F_c = \Lambda(q)w - \Lambda(q)\dot{J}\dot{q} \quad (5.37)$$

with w given by (5.27), the resolved acceleration control scheme can be implemented without the computation of J^{-1}

- Without inertia shaping ($A = \Lambda(q)$) no force feedback is needed (the terms with F_e in τ_c are canceled)

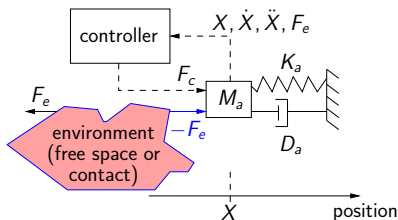
Impedance control XI

Remarks

- ▶ The task space Jacobian J might be a truncated version of the full Jacobian, being described only for task space dimensions
- ▶ The terms $v(\dot{q}, q)$ and $g(q)$ can be represented and compensated in the task space, avoiding joint space pre-compensation (see (5.31)). However, this solution does not compensate the effects of $v(\dot{q}, q)$ and $g(q)$ in all robot space
- ▶ The inertia matrix of the task space $\Lambda(q)$ cannot be computed in task space singularities (although its inverse always exists)
- ▶ Impedance control, in its various forms, deals with robot dynamic behavior in contact, without addressing the servo problem. Some authors (e.g., Siciliano and Vilani [SV99]) call these schemes indirect force control, where position errors are related to contact forces by a mechanical impedance

Impedance control XII

Example: Impedance reshaping for a 1-DOF system [Yos90]



- ▶ 1-DOF mass-spring-damper system with parameters M_a , K_a and D_a , respectively, applying the force F_e on the environment. X is the displacement from the equilibrium of $F_e = F_c = 0$
- ▶ The controller goal is to accomplish a desired mass-spring-damper behavior with parameters M_d , K_d and D_d , respectively

Impedance control XIII

- ▶ The system dynamics is given by

$$M_a\ddot{X} + D_a\dot{X} + K_aX = F_c - F_e \quad (5.38)$$

- ▶ Desired system dynamics is given by

$$M_d\ddot{X} + D_d(\dot{X} - \dot{X}_d) + K_d(X - X_d) = -F_e \quad (5.39)$$

where X_d is the desired position trajectory

- ▶ If X , $\dot{X}$ and $\ddot{X}$ are measurable, F_c can be computed by

$$\begin{aligned} F_c = & (M_a - M_d)\ddot{X} + (D_a - D_d)\dot{X} \\ & + (K_a - K_d)X + D_d\dot{X}_d + K_dX_d \end{aligned} \quad (5.40)$$

Impedance control XIV

- ▶ If F_e is measurable, there is no need to measure $\ddot{X}$ which is very noisy in practice. Isolating $\ddot{X}$ in both equations (5.38)-(5.39), and subtracting them, we get

$$\begin{aligned} F_c = & (1 - M_a/M_d)F_e + (D_a - D_d M_a/M_d)\dot{X} \\ & + (K_a - K_d M_a/M_d)X \\ & + (D_d \dot{X}_d + K_d X_d)M_a/M_d \end{aligned} \quad (5.41)$$

- ▶ If $M_a = M_d$ (i.e., no inertia shaping),

$$F_c = (D_a - D_d)\dot{X} + (K_a - K_d)X + D_d \dot{X}_d + K_d X_d \quad (5.42)$$

which corresponds to a simple PD controller

Impedance control XV

- ▶ The tracking dynamics in free space can be inferred by (5.39), for $F_e = 0$. The transfer function is given by

$$\frac{X(s)}{X_d(s)} = \frac{K_d + D_d s}{M_d s^2 + D_d s + K_d} \quad (5.43)$$

- ▶ The natural frequency

$$W_n = \sqrt{K_d / M_d} \quad (5.44)$$

should be large to improve transient response

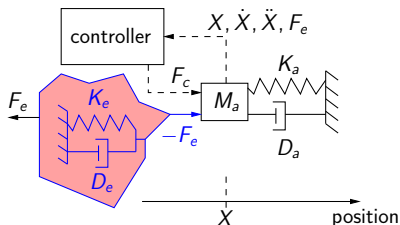
- ▶ The damping coefficient

$$\zeta = \frac{D_d}{2\sqrt{M_d K_d}} \quad (5.45)$$

should be $\zeta \geq 0.7$ to avoid sluggish responses and $\zeta \leq 1.0$ to avoid overshooting

Impedance control XVI

Example: Impedance reshaping for a 1-DOF system with contact parameter knowledge



- In contact the system dynamics becomes

$$M_d \ddot{X} + D_d (\dot{X} - \dot{X}_d) + K_d (X - X_d) = -K_e (X - X_0) - D_e \dot{X} \quad (5.46)$$

where X_0 represents the initial environment contact position

Impedance control XVII

- ▶ Therefore, the tracking dynamics in contact is different from the free space case. It is given by (5.44) and (5.45), changing $D_d \rightarrow D_d + D_e$ and $K_d \rightarrow K_d + K_e$, i.e.,

$$W_n = \sqrt{(K_d + K_e)/M_d} \quad (5.47)$$

and

$$\zeta = \frac{D_d + D_e}{2\sqrt{M_d(K_d + K_e)}} \quad (5.48)$$

- ▶ Knowing K_e and D_e , the impedance parameters M_d , D_d and K_d can be designed to achieve appropriate tracking dynamics. However, contact parameters are typically unknown
- ▶ If K_e is bigger than expected and/or if D_e is smaller than expected, tracking dynamics can be inadequate (i.e., high interaction forces may appear)

Impedance control XVIII

- ▶ Let's consider a numerical example with $M_a = 1$, $D_a = 0.1$ and $K_a = 0$. For the contact parameters, we assume that

$$0 \leq D_e \leq 15 \quad (5.49)$$

and

$$500 \leq K_e \leq 2000 \quad (5.50)$$

- ▶ A desired impedance should be designed such that

$$30 \leq W_n \leq 50 \quad (5.51)$$

and

$$\zeta \geq 0.5 \quad (5.52)$$

Additionally, we want $M_d = 1$ (i.e., no inertia shaping) and K_d and D_d as small as possible

Impedance control XIX

- From (5.47) we get

$$900 \leq K_d + K_e \leq 2500 \quad (5.53)$$

From (5.48) we get

$$D_d \geq \sqrt{K_d + K_e} - D_e \quad (5.54)$$

- The minimum value that satisfies (5.53) is

$$K_d = 400 \quad (5.55)$$

and the smallest D_e that satisfies (5.54) is

$$D_d = \sqrt{2400} \approx 49 \quad (5.56)$$

- From (5.48), the damping coefficient during contact is within

$$0.5 \leq \zeta \leq 1.07 \quad (5.57)$$

Admittance control I

- ▶ The desired system dynamics (5.24) can also be achieved with an admittance control scheme. The admittance controller has as input F_e and outputs a reference position X_r (an impedance controller is the other way around: has position-based inputs and outputs a force)
- ▶ The admittance controller is given by

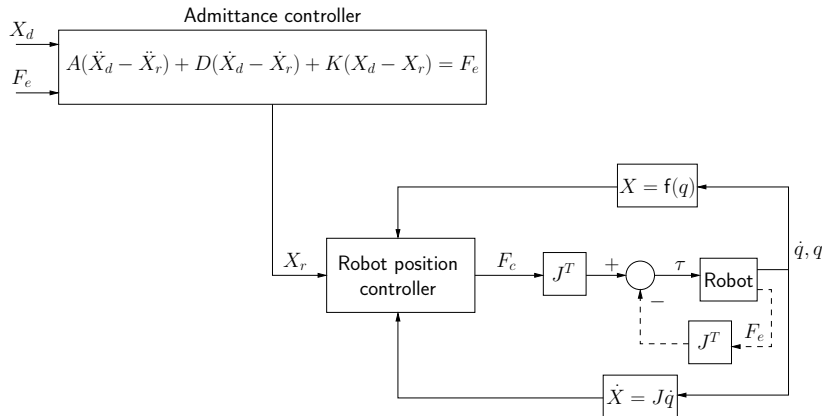
$$A(\ddot{X}_d - \ddot{X}_r) + D(\dot{X}_d - \dot{X}_r) + K(X_d - X_r) = F_e \quad (5.58)$$

where the output of the admittance controller X_r is the reference input to a robot position controller

- ▶ The robot position controller needs to have good tracking capabilities (i.e., $X \approx X_r$) so that the overall system dynamics converges to (5.24). Typically, a PD control with high gains does the job

Admittance control II

Admittance control scheme



- The admittance control scheme has a cascade architecture, where the admittance controller generates references for the inner robot position controller

Admittance control vs. impedance control

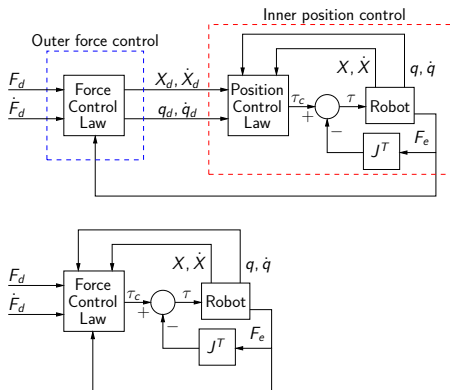
- ▶ Admittance control performs impedance control through an admittance controller
- ▶ Control performance can be boosted by online parameter adaptation based on real-time estimation of environment parameters (e.g., to achieve haptic telepresence, robot impedance needs to be designed as a function of environment impedance)
- ▶ The estimation of environment parameters is a tough problem, though
- ▶ Both impedance and admittance control schemes can also be applied in the joint space. In this case, q plays the role of X , and only joint space dynamics is needed

Explicit force control I

- ▶ It involves direct control of the force exerted on the environment (F_e)
- ▶ Servo (i.e., command following) specifications help to design the control architecture
- ▶ Two types of explicit force control have been proposed
 - ▶ Force based, employing some form of PID control
 - ▶ Inner position/outer force loop based. The outer force control loop provides position (or velocity) commands to the inner loop
- ▶ Volpe and Khosla [VK93] have analyzed explicit force control schemes, showing trade-offs between force tracking performance (high control gains) and stability

Explicit force control II

General control schemes for explicit force control

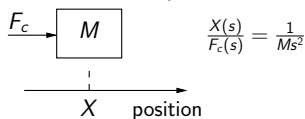


- Both schemes have force based signals as reference inputs (F_d and $\dot{F}_d$), instead of position based ones. This is the main difference with respect to impedance control schemes

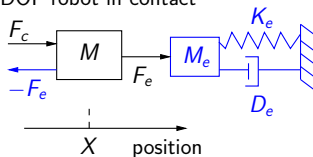
Explicit force control III

Example: Coupled dynamics problem for 1-DOF robot

1-DOF robot in free space



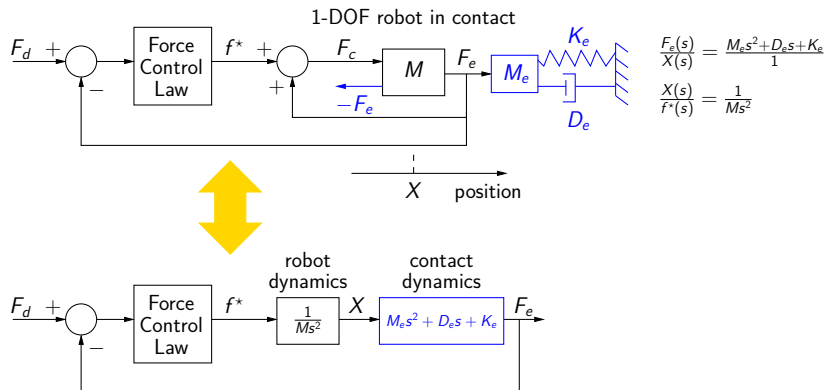
1-DOF robot in contact



- Contact changes the overall system dynamics, creating difficulties to dynamic control since contact parameters are poorly known

Explicit force control IV

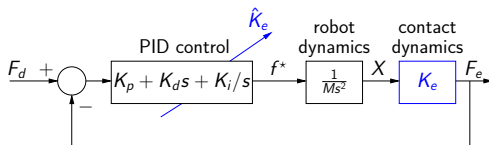
Force control with dynamic decoupling for a 1-DOF robot



- Feeding back the applied force F_e with positive feedback makes the control variable f^* to "see" robot dynamics decoupled from contact dynamics

Explicit force control V

Example: PID Force control with dynamic decoupling for a 1-DOF robot

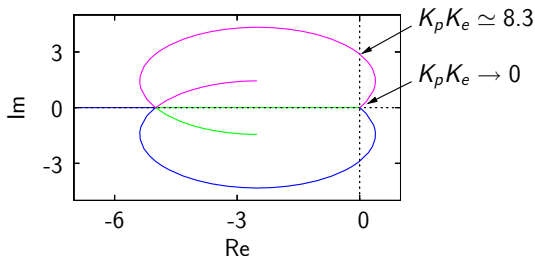


- ▶ Adaptive force control to achieve dynamic consistency between F_d and F_e , if contact parameters are known
- ▶ The characteristic polynomial $\Delta(s) = Ms^3 + K_d K_e s^2 + K_p K_e s + K_i K_e$
- ▶ Making $\Delta(s) = M(s + w_j)^3$, the solution is then (compare it with (4.8))

$$K_p = 3Mw_j^2/\hat{K}_e \quad K_d = 3Mw_j/\hat{K}_e \quad K_i = Mw_j^3/\hat{K}_e \quad (5.59)$$

Explicit force control VI

Root locus



- In this example, $M = 1$, $w_j = 5$, and the PID control law is given by (see (5.59))

$$K_d = \frac{K_p}{w_j} \quad K_i = \frac{K_p w_j}{3} \quad (5.60)$$

Explicit force control VII

- ▶ The root locus shows that for any fixed K_p , the system is stable only if

$$K_e > 8.3/K_p \quad (5.61)$$

- ▶ K_p is given by (see (5.59))

$$K_p = 75/\hat{K}_e \quad (5.62)$$

- ▶ Therefore, writing

$$\hat{K}_e = \alpha_e K_e \quad (5.63)$$

where $\alpha_e > 0$ reflects the stiffness estimation mismatch, stability is assured if (see (5.61), (5.62) and (5.63))

$$\alpha_e < 9.03 \quad (5.64)$$

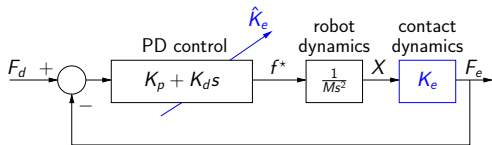
Explicit force control VIII

Remarks

- ▶ Adaptive PID control requires estimating the contact stiffness K_e
- ▶ For $K_e \rightarrow 0$, the PID gains of (5.59) go to infinity, creating practical difficulties. A lower-bound limit in $\hat{K}_e$ avoids computational problems, but it can generate instability (see (5.63) and (5.64))
- ▶ In contact, stability is only guaranteed if the PID gains satisfy conditions
- ▶ PD control can be used to avoid stability problems

Explicit force control IX

Example: PD Force control with dynamic decoupling for a 1-DOF robot

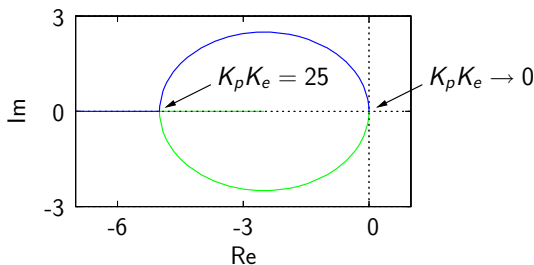


- ▶ The characteristic polynomial $\Delta(s) = Ms^2 + K_d K_e s + K_p K_e$
- ▶ Making $\Delta(s) = M(s + w_j)^2$, the solution is then

$$K_p = 2Mw_j / \hat{K}_e \quad K_d = Mw_j^2 / \hat{K}_e \quad (5.65)$$

Explicit force control X

Root locus



- In this example, $M = 1$, $w_j = 5$, and the PD control law is given by (see (5.65))

$$K_d = 2 \frac{K_p}{w_j} \quad (5.66)$$

Explicit force control XI

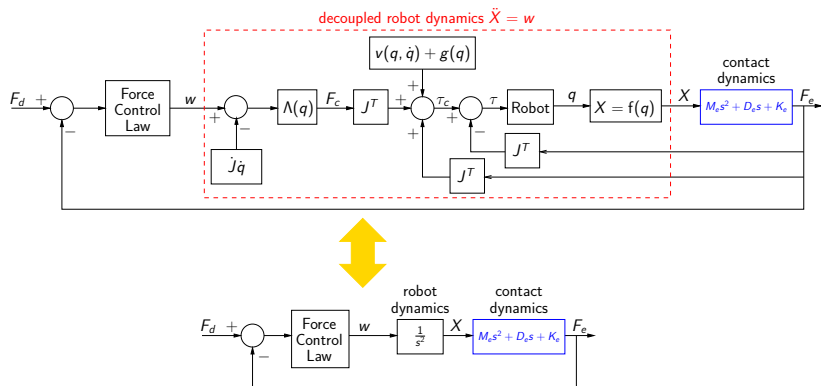
- ▶ The root locus shows that the system is always stable, except for the free space condition

$$K_p K_e = 0 \quad (5.67)$$

- ▶ To avoid the effect of a zero in the transfer function (which affects the response dynamics), the PD control can be reformulated by damping only the output $-K_d s F_e$, instead of the error $K_d s (F_d - F_e)$

Explicit force control XII

Robot force control with dynamic decoupling



- The force control law "sees" for each Cartesian dimension a unitary mass interacting with the environment

Chapter 6

Kalman active observers

This chapter introduces Kalman active observers (AOBs), which have been applied in several force and motion control architectures, such as explicit force control of robot manipulators [CKNH00], [CKNH01], [PCK04], haptic telemanipulation [CPK06], multi-contact force control in humanoids [PK05], and motion control in mobile robotics [CN05], [BCH05], [CB04], [MCN⁺03]. This chapter is based on [Cor07].

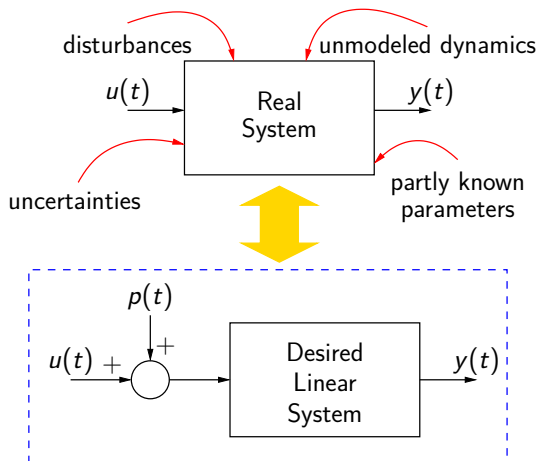
Introduction

- ▶ In complex control systems, unknown disturbances, higher order dynamics, nonlinearities and noise are always present
- ▶ The effect of joint measurement noise on robot orientation can be seen in [CS21]
- ▶ To tackle the disturbance decoupling problem, PID-based techniques [EM99], state feedback [CM00] geometric concepts [CDH97], tracking schemes [CLL02] and observer techniques [Oda01] have been proposed among others
- ▶ In the AOB, disturbance estimation is modeled as an auto-regressive (AR) process with fixed parameters driven by a random source. This process represents stochastic evolutions
- ▶ The AOB provides a methodology to achieve model-reference adaptive control through extra states and stochastic design

AOB concept I

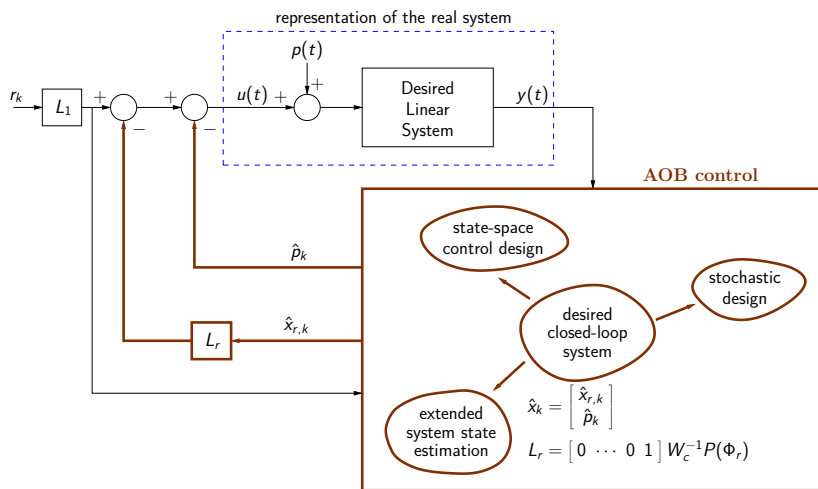
- ▶ A linear system represented in state space can be controlled through state feedback (e.g. optimal control, adaptive control and explicit pole placement control). In practice, the main problem of this approach is that unmodeled terms including unknown disturbances are not addressed in the control design
- ▶ Therefore, it is necessary to develop a control structure that can deal with them, to boost control performance
- ▶ The main goal of the AOB is to achieve model-reference adaptive control using Kalman techniques. The AOB is based on:
 1. A desired closed loop system
 2. An extra equation to estimate an equivalent disturbance referred to the system input. An active state p_k (extra-state) is introduced to compensate unmodeled terms, providing a feedforward compensation action
 3. The stochastic design of the Kalman matrices Q_k (model uncertainty) and R_k (measure uncertainty) for the AOB context

AOB concept II



- Representation of the real system by a nominal linear system plus an equivalent disturbance $p(t)$ referred to the system input

AOB concept III



- ▶ AOB control scheme, with extended state feedback
- ▶ Model reference adaptive control based on Kalman techniques

AOB concept IV

- Given a system with equations

$$x_{r,k} = \Phi_r x_{r,k-1} + \Gamma_r u_{k-1} + \xi_{x_{r,k}} \quad (6.1)$$

and

$$y_k = C_r x_{r,k} + \eta_k, \quad (6.2)$$

an observer of the state $x_{r,k}$, $\hat{x}_{r,k}$, can be written as

$$\begin{aligned} \hat{x}_{r,k} = & \Phi_{r,n} \hat{x}_{r,k-1} + \Gamma_{r,n} u_{k-1} \\ & + K_k [y_k - C_r (\Phi_{r,n} \hat{x}_{r,k-1} + \Gamma_{r,n} u_{k-1})], \end{aligned} \quad (6.3)$$

where $\Phi_{r,n}$ and $\Gamma_{r,n}$ are respectively the nominal state transition and command matrices (i.e., the ones used in the design)

AOB concept V

- ▶ Φ_r and Γ_r are the real matrices. $\xi_{x_{r,k}}$ and η_k are Gaussian random variables associated to the system and measures, respectively, having a key role in the AOB design
- ▶ Defining the estimation error as

$$e_{r,k} = x_{r,k} - \hat{x}_{r,k}, \quad (6.4)$$

and considering ideal conditions (i.e., the nominal matrices are equal to the real ones and $\xi_{x_{r,k}}$ and η_k are zero), $e_{r,k}$ can be computed from (6.1) and (6.3). Its value is

$$e_{r,k} = (\Phi_r - K_k C_r \Phi_r) e_{r,k-1} \quad (6.5)$$

- ▶ The error dynamics given by the eigenvalues of $(\Phi_r - K_k C_r \Phi_r)$ is function of the K_k gain. The Kalman observer computes the best K_k in a straightforward way, minimizing the mean square error of the state estimate due to the random sources $\xi_{x_{r,k}}$ and η_k .

AOB concept VI

- ▶ When there are unmodeled terms, (6.5) needs to be changed. A deterministic description of $e_{r,k}$ is difficult, particularly when unknown modeling errors exist. Hence, a stochastic approach is attempted to describe it
- ▶ If state feedback from the observer is used to control the system, p_k enters as an additional input

$$p_k = -L_r e_{r,k}, \quad (6.6)$$

where L_r is the state feedback gain

- ▶ A state space equation should be found to characterize this undesired input, leading the system to an extended state representation
- ▶ The active state¹ $\hat{p}_k$ estimates the extended state p_k
permitting the overall system to have the desired behavior

¹The name "active state" is used for both p_k and $\hat{p}_k$. p_k can be seen as an equivalent disturbance state that exists in the real system, and $\hat{p}_k$ is the estimation of p_k .

Active state equations I

- ▶ The p_k equation has a key role in the AOB structure, crossing the AOB design.
- ▶ To be able to track functions with unknown dynamics, a stochastic equation is used to describe p_k ,

$$p_k - p_{k-1} = \xi_{p_k}, \quad (6.7)$$

in which ξ_{p_k} is a zero-mean Gaussian random variable²

- ▶ Equation (6.7) says that the first derivative (or first-order evolution) of p_k is randomly distributed

²The mathematical notation is for single input systems. For multiple input systems, p_k in (6.7) is a column vector with dimension equal to the number of inputs.

Active state equations II

- ▶ Defining ${}^N\xi_{p_k}$ as the N^{th} -order evolution of ξ_{p_k} (or the $(N+1)^{th}$ -order evolution of p_k),

$${}^N\xi_{p_k} = {}^{N-1}\xi_{p_k} - {}^{N-1}\xi_{p_{k-1}}, \quad \text{with } {}^0\xi_{p_k} = \xi_{p_k}, \quad (6.8)$$

the general form of (6.7) is

$$p_k = \sum_{j=1}^N (-1)^{j+1} \frac{N!}{j!(N-j)!} p_{k-j} + {}^{N-1}\xi_{p_k} \quad (6.9)$$

- ▶ p_k has fixed parameters given by (6.9) and is driven by the statistics of ${}^{N-1}\xi_{p_k}$
- ▶ The stochastic equations (6.7) for the AOB-1 or (6.9) for the AOB-N are used to describe p_k
- ▶ If ${}^{N-1}\xi_{p_k} = 0$, (6.9) is a deterministic model for any disturbance p_k that has its N^{th} -derivative equal to zero. In this way, the stochastic information present in ${}^{N-1}\xi_{p_k}$ gives more flexibility to p_k , since its evolutionary model is not rigid

State space discretization I

- Discretizing the system plant (without considering disturbances)

$$\begin{cases} \frac{dx_r(t)}{dt} &= Ax_r(t) + Bu(t - \tau) \\ y(t) &= C_r x_r(t) \end{cases} \quad (6.10)$$

with sampling time h and dead-time

$$\tau = (d - 1)h + \tau', \quad \text{with } 0 < \tau' \leq h, \quad (6.11)$$

State space discretization II

- The following discrete time system is obtained

$$\begin{bmatrix} x'_k \\ u_{k-d} \\ \vdots \\ u_{k-2} \\ u_{k-1} \end{bmatrix} = \begin{bmatrix} \Phi_1 & \Gamma_1 & \Gamma_0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x'_{k-1} \\ u_{k-d-1} \\ \vdots \\ u_{k-3} \\ u_{k-2} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u_{k-1} \quad (6.12)$$

and

$$y_k = C_r x_{r,k} \quad (6.13)$$

- The state $x_{r,k}$ is

$$x_{r,k} = \begin{bmatrix} x'_k & u_{k-d} & \cdots & u_{k-2} & u_{k-1} \end{bmatrix}^T, \quad (6.14)$$

in which x'_k is the system state considering no dead-time.
Therefore, τ of (6.10) increases the system order

State space discretization III

- Φ_1 , Γ_0 and Γ_1 are given by (6.15) to (6.17), respectively [ÅW97]

$$\Phi_1 = e^{Ah} = \phi(h), \quad (6.15)$$

$$\Gamma_0 = \int_0^{h-\tau'} \phi(\lambda) d\lambda B \quad (6.16)$$

and

$$\Gamma_1 = \phi(h - \tau') \int_0^{\tau'} \phi(\lambda) d\lambda B \quad (6.17)$$

AOB-1 algorithm I

- The augmented state space representation is

$$\begin{bmatrix} x_{r,k} \\ p_k \end{bmatrix} = \begin{bmatrix} \Phi_r & \Gamma_r \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ p_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} u'_{k-1} + \begin{bmatrix} \xi_{x_{r,k}} \\ \xi_{p_k} \end{bmatrix}, \quad (6.18)$$

where

$$u'_{k-1} = r_{k-1} - \begin{bmatrix} L_r & 1 \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ \hat{p}_{k-1} \end{bmatrix}, \quad (6.19)$$

$$\Phi_r = \begin{bmatrix} \Phi_1 & \Gamma_1 & \Gamma_0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \quad (6.20)$$

and

$$\Gamma_r = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix}^T \quad (6.21)$$

AOB-1 algorithm II

- L_r is obtained by any control technique applied to (6.12) to achieve a desired closed loop behavior. The measurement equation is

$$y_k = C \begin{bmatrix} x_{r,k} \\ p_k \end{bmatrix} + \eta_k, \quad (6.22)$$

with

$$C = \begin{bmatrix} C_r & 0 \end{bmatrix} \quad (6.23)$$

- The desired closed loop system appears when $\hat{p}_k = p_k$, i.e.,

$$\begin{bmatrix} x_{r,k} \\ p_k \end{bmatrix} = \begin{bmatrix} \Phi_r - \Gamma_r L_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ p_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} r_{k-1} + \begin{bmatrix} \xi_{x_{r,k}} \\ \xi_{p_k} \end{bmatrix} \quad (6.24)$$

AOB-1 algorithm III

- ▶ The state $x_{r,k}$ in (6.24) is accurate if most of the modeling errors are merged to p_k . Hence, $\xi_{x_{r,k}}$ should be small compared to ξ_{p_k}
- ▶ The state estimation must consider not only the influence of the uncertainty $\xi_{x_{r,k}}$, but also the deterministic term due to the reference input, the extended state representation and the desired closed loop response. It is given by³

$$\begin{aligned} \begin{bmatrix} \hat{x}_{r,k} \\ \hat{p}_k \end{bmatrix} &= \begin{bmatrix} \Phi_{r,n} - \Gamma_{r,n}L_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_{r,k-1} \\ \hat{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{r,n} \\ 0 \end{bmatrix} r_{k-1} \\ &+ K_k \left\{ y_k - C \left(\begin{bmatrix} \Phi_{r,n} - \Gamma_{r,n}L_r & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_{r,k-1} \\ \hat{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{r,n} \\ 0 \end{bmatrix} r_{k-1} \right) \right\} \end{aligned} \quad (6.25)$$

³ $\Phi_{r,n}$ and $\Gamma_{r,n}$ represent the nominal values of Φ_r and Γ_r , respectively.

AOB-1 algorithm IV

- ▶ The Kalman gain K_k is given by

$$K_k = P_{1k} C^T [C P_{1k} C^T + R_k]^{-1}, \quad (6.26)$$

with

$$P_{1k} = \Phi_n P_{k-1} \Phi_n^T + Q_k, \quad (6.27)$$

and

$$P_k = P_{1k} - K_k C P_{1k} \quad (6.28)$$

- ▶ Φ_n is the augmented open loop matrix used in the design,

$$\Phi_n = \begin{bmatrix} \Phi_{r,n} & \Gamma_{r,n} \\ 0 & 1 \end{bmatrix} \quad (6.29)$$

- ▶ Q_k is the system noise matrix

$$Q_k = \begin{bmatrix} Q_{x_{r,k}} & 0 \\ 0 & Q_{p_k} \end{bmatrix} \quad (6.30)$$

- ▶ R_k is the measurement noise matrix, $R_k = E \{ \eta_k \eta_k^T \}$, and P_k is the mean square error matrix

AOB-N algorithm I

- ▶ The AOB-1 algorithm has to be slightly changed for the AOB-N. Only the equation of the active state changes, entailing minor modifications in the overall AOB design
- ▶ Equation (6.9) has the following state space representation

$$\begin{bmatrix} p_{k-(N-1)} \\ p_{k-(N-2)} \\ \vdots \\ p_{k-1} \\ p_k \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_N & a_{N-1} & a_{N-2} & \cdots & a_1 \end{bmatrix} \begin{bmatrix} p_{k-N} \\ p_{k-(N-1)} \\ \vdots \\ p_{k-2} \\ p_{k-1} \end{bmatrix} \quad (6.31)$$

$$+ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ {}^{N-1}\zeta_{p_k} \end{bmatrix}, \text{ with } a_i = (-1)^{i+1} \frac{N!}{i!(N-i)!}, \quad i = 1, \dots, N$$

AOB-N algorithm II

- ▶ In compact form, (6.31) is represented by

$${}^N p_k = \Phi_{2,2} {}^N p_{k-1} + \xi_{{}^N p_k} \quad (6.32)$$

- ▶ Equation (6.18) can be re-written as

$$\begin{bmatrix} x_{r,k} \\ {}^N p_k \end{bmatrix} = \begin{bmatrix} \Phi_r & \Phi_{1,2} \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ {}^N p_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} u'_{k-1} + \begin{bmatrix} \xi_{x_{r,k}} \\ \xi_{{}^N p_k} \end{bmatrix}, \quad (6.33)$$

where

$${}^N p_k = \begin{bmatrix} p_{k-(N-1)} & p_{k-(N-2)} & \cdots & p_{k-1} & p_k \end{bmatrix}^T \quad (6.34)$$

AOB-N algorithm III

- $\Phi_{2,2}$ is the state transition matrix of (6.31),

$$\Phi_{1,2} = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \quad (6.35)$$

and

$$\xi_{Np_k} = \begin{bmatrix} 0 & 0 & \cdots & 0 & {}^{N-1}\xi_{p_k} \end{bmatrix}^T \quad (6.36)$$

- The desired closed loop of (6.24) is changed to

$$\begin{bmatrix} x_{r,k} \\ {}^Np_k \end{bmatrix} = \begin{bmatrix} \Phi_L & 0 \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} x_{r,k-1} \\ {}^Np_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_r \\ 0 \end{bmatrix} r_{k-1} + \begin{bmatrix} \xi_{x_{r,k}} \\ \xi_{Np_k} \end{bmatrix} \quad (6.37)$$

AOB-N algorithm IV

- Φ_L is given by

$$\Phi_L = \begin{bmatrix} \Phi_1 & \Gamma_1 & \Gamma_0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -L_1 & -L_2 & -L_3 & \cdots & -L_M \end{bmatrix} \quad (6.38)$$

- The L_r components ($L_1, \dots, L_M$) can be obtained by Ackermann's formula, i.e.,

$$L_r = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix} W_c^{-1} P(\Phi_r). \quad (6.39)$$

W_c is the reachability matrix

$$W_c = \begin{bmatrix} \Gamma_r & \Phi_r \Gamma_r & \cdots & \Phi_r^{M-1} \Gamma_r \end{bmatrix}, \quad (6.40)$$

and $P(\Phi_r)$ is the desired characteristic polynomial

AOB-N algorithm V

- ▶ The state estimation is⁴

$$\begin{aligned} \begin{bmatrix} \hat{x}_{r,k} \\ N\hat{p}_k \end{bmatrix} &= \begin{bmatrix} \Phi_{L,n} & 0 \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} \hat{x}_{r,k-1} \\ N\hat{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{r,n} \\ 0 \end{bmatrix} r_{k-1} \\ &+ K_k \left\{ y_k - C \left(\begin{bmatrix} \Phi_{L,n} & 0 \\ 0 & \Phi_{2,2} \end{bmatrix} \begin{bmatrix} \hat{x}_{r,k-1} \\ N\hat{p}_{k-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{r,n} \\ 0 \end{bmatrix} r_{k-1} \right) \right\} \end{aligned} \quad (6.41)$$

- ▶ K_k is given by (6.26) to (6.28), with

$$\Phi_n = \begin{bmatrix} \Phi_{r,n} & \Phi_{1,2} \\ 0 & \Phi_{2,2} \end{bmatrix} \quad (6.42)$$

⁴ $\Phi_{L,n}$ is the nominal value of Φ_L .

AOB-N algorithm VI

- ▶ The state feedback gain L is

$$L = \begin{bmatrix} L_r & L_{Np_k} \end{bmatrix}, \quad (6.43)$$

with

$$L_{Np_k} = \begin{bmatrix} 0 & \cdots & 0 & 1 \end{bmatrix} \quad (6.44)$$

AOB estimation strategies I

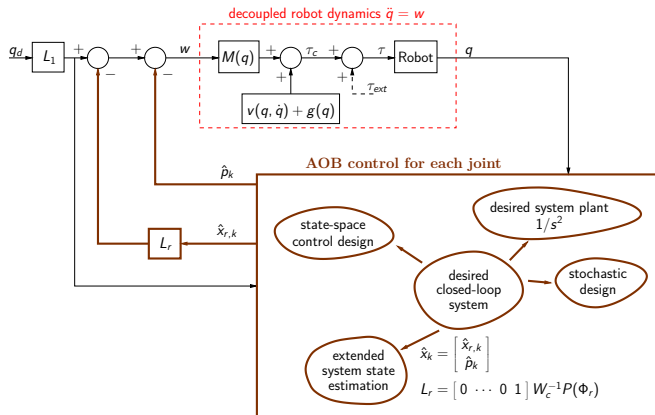
- ▶ Multiplying the matrices Q_k , R_k and P_{-1} by the same positive factor entails the same Kalman gain [CKNH01]
- ▶ Therefore, the estimation strategy only depends on the relations between the Q_k values, the R_k values, and both Q_k and R_k values
- ▶ In the AOB, the relations between the Q_k values are straightforward

AOB estimation strategies II

- ▶ If model accuracy is very good compared to measure accuracy, a model-based approach (MBA) is followed. The estimation is mainly based on model information giving little importance to measures. The Kalman gain has low values
- ▶ On the other hand, if the measures are very accurate, a sensor-based approach (SBA) is followed. The Kalman gain has high values
- ▶ The hybrid-based approach (HBA) is the general form of the AOB and establishes a trade-off between SBA and MBA, i.e., it balances the estimates based on sensory and model information

Robot control with AOBs I

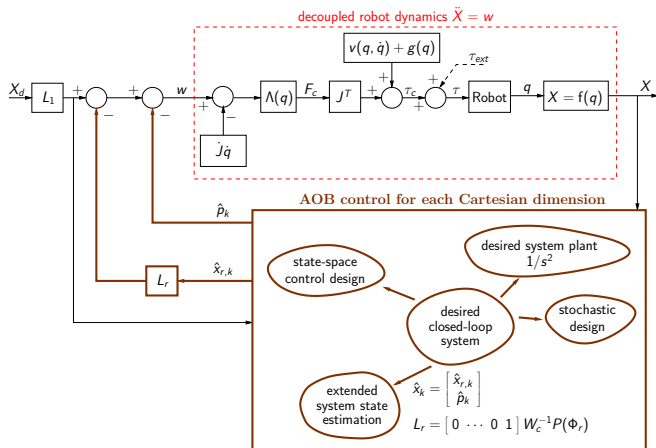
Joint space motion control



- The AOB control imposes a desired tracking dynamics in the joint space, in spite of system disturbances

Robot control with AOBs II

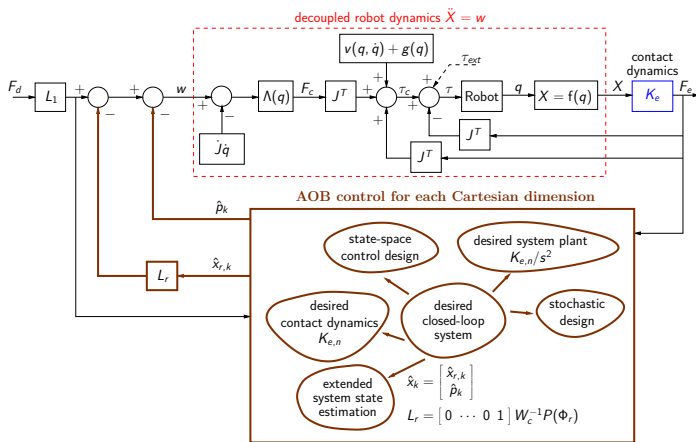
Task space motion control



- ▶ The AOB control imposes a desired tracking dynamics in the task space, in spite of system disturbances

Robot control with AOBs III

Task space force control with desired stiffness $K_{e,n}$



- The AOB control imposes a desired force tracking dynamics, in spite of system disturbances

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